Regression Analysis (Linear Model) Part A

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From ANOVA to Regression models

- **Anova Models**
  - For answering the following questions:
    - Does a subset of Factors affect the response variable?
    - What is the influence **structure** of the Factors?
    - How to select the Factor optimal values **among** the **experimented ones (treatments)**?
  - Topics about Designing Experiments
    - How to Design an experiment with a given **Power** and **Alfa**
    - How to reduce the Total Number of Experiments **N**
    - How to Design experiments blocking the Nuisance Factor Effects (if known)

Regression Models

– **Prediction**:

✓**Point prediction** and related topics

✓**Confidence Interval (CI)** about the expected value of the response variable

✓**Prediction Confidence Interval** (**PI**) about the single realization of the response variable

– **Building a model**

✓Model significance

✓Adding or skipping model terms

Random Vectors

$$\boldsymbol{y} = \begin{Bmatrix} y_1 \\ y_2 \\ . \\ . \\ y_N \end{Bmatrix} \quad \text{with} \quad y_i \sim \mathrm{Dist}(\mu_i, \sigma_i^2) \qquad \boldsymbol{E}(\boldsymbol{y}) = \begin{Bmatrix} E(y_1) \\ E(y_2) \\ . \\ . \\ E(y_N) \end{Bmatrix}$$

$$\boldsymbol{\Sigma}_{\mathbf{y}} = E[(\mathbf{y} - E(\mathbf{y}))(\mathbf{y} - E(\mathbf{y}))^T] = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & . & . & \sigma_{1N} \\ \sigma_{12} & \sigma_2^2 & . & . & \sigma_{2N} \\ . & . & . & . & . \\ . & . & . & . & . \\ \sigma_{1N} & \sigma_{2N} & . & . & \sigma_N^2 \end{bmatrix}$$

Let us consider

$\mathbf{a}^T\mathbf{y}$

$$V\left(\mathbf{a}^T\mathbf{y}\right) = \boldsymbol{\Sigma}_{\mathbf{a}^T\mathbf{y}} = \mathbf{a}^{\mathrm{T}}\boldsymbol{\Sigma}_{\mathbf{y}}\mathbf{a}$$

e.g. $y = a_1 y_1 + a_2 y_2 + a_3 y_3$ **and** y_i **independent**

$$\boldsymbol{a} = \begin{Bmatrix} a_1 \\ a_2 \\ a_3 \end{Bmatrix} \qquad \boldsymbol{y} = \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \end{Bmatrix} \qquad \boldsymbol{\Sigma}_y = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{bmatrix}$$

$$\boldsymbol{\Sigma}_{\boldsymbol{a}^T\boldsymbol{y}} = \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \\ a_3 \end{Bmatrix} = \sum_{i=1}^{3} a_i^2 \sigma_i^2$$

Try with off-diagonal elements different from zero

Covariance of two Random vectors $\mathbf{h}$ (p x 1) e $\mathbf{z}$ (q x 1)

$$\mathbf{Cov}\left(\mathbf{h},\mathbf{z}\right)=\mathbf{\Sigma}_{\mathbf{h},\mathbf{z}}=E\left[\left(\mathbf{h}-E\left(\mathbf{h}\right)\right)\left(\mathbf{z}-E\left(\mathbf{z}\right)\right)^{T}\right]$$

If $\mathbf{h}$ is (p x 1) and $\mathbf{z}$ is (q x 1) the **covariance matrix** is (p x q)

Let us consider the two linear combinations

$$\mathbf{a}^{\mathrm{T}}\mathbf{y} \text{ and } \mathbf{b}^{\mathrm{T}}\mathbf{y}$$

$$\Downarrow$$

$$COV(\mathbf{a}^{\mathrm{T}}\mathbf{y}\,,\ \mathbf{b}^{\mathrm{T}}\mathbf{y}) = \boldsymbol{\Sigma}_{\mathbf{a}^{\mathrm{T}}\mathbf{y},\mathbf{b}^{\mathrm{T}}\mathbf{y}} = \mathbf{a}^{\mathrm{T}}\boldsymbol{\Sigma}_{\mathbf{y}}\mathbf{b}$$

Let us consider the following Matrices/Vectors

$$\boldsymbol{y}\ (N*1) \quad \boldsymbol{A} \quad (r*N) \quad \boldsymbol{B} \quad (s*N)$$

$$\boldsymbol{h} = \boldsymbol{A}\boldsymbol{y} \qquad \boldsymbol{z} = \boldsymbol{B}\boldsymbol{y}$$

$$\boldsymbol{\Sigma}_{\boldsymbol{h}} = \boldsymbol{A}\boldsymbol{\Sigma}_{\boldsymbol{y}}\boldsymbol{A}^{T} \qquad \boldsymbol{\Sigma}_{\boldsymbol{z}} = \boldsymbol{B}\boldsymbol{\Sigma}_{\boldsymbol{y}}\boldsymbol{B}^{T} \qquad \boldsymbol{\Sigma}_{\boldsymbol{h},\boldsymbol{z}} = \boldsymbol{A}\boldsymbol{\Sigma}_{\boldsymbol{y}}\boldsymbol{B}^{T}$$

Proofs

$$\mathbf{V}\left(\mathbf{a}^T\mathbf{y}\right) = \boldsymbol{\Sigma}_{\mathbf{a}^T\mathbf{y}} = E\left(\left(\mathbf{a}^T\mathbf{y} - E\left(\mathbf{a}^T\mathbf{y}\right)\right)\left(\mathbf{a}^T\mathbf{y} - E\left(\mathbf{a}^T\mathbf{y}\right)\right)^T\right) =$$

$$= \mathbf{a}^{\mathrm{T}} E\left(\left(\mathbf{y} - E\left(\mathbf{y}\right)\right)\left(\mathbf{y} - E\left(\mathbf{y}\right)\right)^T\right)\mathbf{a} = \mathbf{a}^{\mathrm{T}}\boldsymbol{\Sigma}_{\mathbf{y}}\mathbf{a}$$

$$\mathbf{COV}(\mathbf{a}^{\mathrm{T}}\mathbf{y}\ ,\ \mathbf{b}^{\mathrm{T}}\mathbf{y}) = \boldsymbol{\Sigma}_{\mathbf{a}^{\mathrm{T}}\mathbf{y},\mathbf{b}^{\mathrm{T}}\mathbf{y}} = E\left(\left(\mathbf{a}^T\mathbf{y} - E\left(\mathbf{a}^T\mathbf{y}\right)\right)\left(\mathbf{b}^T\mathbf{y} - E\left(\mathbf{b}^T\mathbf{y}\right)\right)^T\right) =$$

$$= \mathbf{a}^T E\left(\left(\mathbf{y} - E\left(\mathbf{y}\right)\right)\left(\mathbf{y} - E\left(\mathbf{y}\right)\right)^T\right)\mathbf{b} = \mathbf{a}^{\mathrm{T}}\boldsymbol{\Sigma}_{\mathbf{y}}\mathbf{b}$$

- **Trace** of a square matrix A

$$Tr(\mathbf{A}) = \sum_i a_{ii}$$

- If **A** e **B** are two matrices and if **AB** e **BA** make sense

$$Tr(\mathbf{AB}) = Tr(\mathbf{BA})$$

- if **A+B** makes sense

$$Tr(\mathbf{A}+\mathbf{B}) = Tr(\mathbf{A}) + Tr(\mathbf{B})$$

Proof

$$Tr\left(\mathbf{AB}\right) = Tr\left(\mathbf{BA}\right)$$

$$\mathbf{A} \rightarrow p * q \quad \mathbf{B} \rightarrow q * p$$

$$\mathbf{AB} \rightarrow p * p \quad \mathbf{BA} \rightarrow q * q$$

$$\mathbf{AB}_{ii} = \sum_{j=1,q} a_{ij} b_{ji}$$

$$Tr\left(\mathbf{AB}\right) = \sum_{i=1,p} \mathbf{AB}_{ii} = \sum_{i=1,p} \sum_{j=1,q} a_{ij} b_{ji} = \sum_{j=1,q} \sum_{i=1,p} b_{ji} a_{ij} = Tr\left(\mathbf{BA}\right)$$

Eigenvalues and Eigenvectors

- Sometime we will use **eigenvalues** and **eigenvectors**.
- We'll do it for **symmetrical (square matrices) and positive definite matrices Σ (pxp)**
- **Positive Definite Matrix** $\mathbf{a}^T \boldsymbol{\Sigma} \mathbf{a} > 0 \quad \forall \mathbf{a} \neq \mathbf{0}$
- **Eigenvalues** $\lambda_i \rightarrow |\boldsymbol{\Sigma} - \lambda \mathbf{I}| = \mathbf{0}$
- **Eigenvalues properties for symmetrical…….**

$$\lambda_i > 0$$

$$|\boldsymbol{\Sigma}| = \lambda_1 \lambda_2 \ldots \lambda_p$$

$$Tr(\boldsymbol{\Sigma}) = \lambda_1 + \lambda_2 + \ldots + \lambda_p$$

- **<u>Normalized Eigenvectors</u>** $\mathbf{m}_i$

$$\mathbf{\Sigma m}_i = \lambda_i \mathbf{m}_i \rightarrow \begin{cases} \mathbf{m}_i^T \mathbf{m}_j = 0 & \forall i \neq j \\ \mathbf{m}_i^T \mathbf{m}_i = 1 \end{cases}$$

- **<u>Spectral Decomposition</u>** of $\mathbf{\Sigma}$ and $\mathbf{\Sigma}^{-1}$

$$\mathbf{\Sigma} = \lambda_1 \mathbf{m}_1 \mathbf{m}_1^T + \lambda_2 \mathbf{m}_2 \mathbf{m}_2^T + \ldots + \lambda_p \mathbf{m}_p \mathbf{m}_p^T$$

$$\mathbf{\Sigma}^{-1} = \frac{1}{\lambda_1} \mathbf{m}_1 \mathbf{m}_1^T + \frac{1}{\lambda_2} \mathbf{m}_2 \mathbf{m}_2^T + \ldots + \frac{1}{\lambda_p} \mathbf{m}_p \mathbf{m}_p^T$$

- **Quadratic Form** $\boxed{\mathbf{y}^T\mathbf{By}}$

$$\mathbf{y}^T\mathbf{By} = \begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \end{Bmatrix} = b_{11}y_1^2 + b_{22}y_2^2 + \left(b_{12} + b_{21}\right) y_1 y_2$$

Remark:

If you want a complete quadratic function with the constant term

$$b_0 + \mathbf{b}^T\mathbf{y} + \mathbf{y}^T\mathbf{By}$$

$$\boxed{\begin{gathered} \text{if } E(\mathbf{y}) = \boldsymbol{\mu} \quad \text{e} \quad V(\mathbf{y}) = \boldsymbol{\Sigma}_{\mathbf{y}} \\ E\left(\mathbf{y}^T\mathbf{By}\right) = \boldsymbol{\mu}^T\mathbf{B}\boldsymbol{\mu} + Tr\left(\mathbf{B}\boldsymbol{\Sigma}_{\mathbf{y}}\right) \end{gathered}}$$

$$\boxed{\begin{gathered} \text{if } \boldsymbol{y} \sim MN\left(\boldsymbol{\mu}, \boldsymbol{\Sigma}_{\boldsymbol{y}}\right) \\ \left(\boldsymbol{y} - \boldsymbol{\mu}\right)^T \boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1} \left(\boldsymbol{y} - \boldsymbol{\mu}\right) \sim \chi^2\left(N\right) \end{gathered}}$$

Proof 1

$$\boxed{\begin{gathered}\text{if } E(\mathbf{y}) = \boldsymbol{\mu} \quad \text{e} \quad V(\mathbf{y}) = \boldsymbol{\Sigma}_{\mathbf{y}} \\ E\left(\mathbf{y}^T \mathbf{B} \mathbf{y}\right) = \boldsymbol{\mu}^T \mathbf{B} \boldsymbol{\mu} + Tr\left(\mathbf{B} \boldsymbol{\Sigma}_{\mathbf{y}}\right)\end{gathered}}$$

$$E\left(\mathbf{y}^T \mathbf{B} \mathbf{y}\right) = E\left(Tr\left(\mathbf{y}^T \mathbf{B} \mathbf{y}\right)\right) = E\left(Tr\left(\mathbf{B} \mathbf{y} \mathbf{y}^T\right)\right) = Tr\left(\mathbf{B} E\left(\mathbf{y} \mathbf{y}^T\right)\right) = *$$

$$\boldsymbol{\Sigma}_{\mathbf{y}} = E\left((\mathbf{y} - \boldsymbol{\mu})(\mathbf{y} - \boldsymbol{\mu})^T\right) = E\left(\mathbf{y}\mathbf{y}^T - \mathbf{y}\boldsymbol{\mu}^T - \boldsymbol{\mu}\mathbf{y}^T + \boldsymbol{\mu}\boldsymbol{\mu}^T\right) = E\left(\mathbf{y}\mathbf{y}^T\right) - \boldsymbol{\mu}\boldsymbol{\mu}^T$$

$$E\left(\mathbf{y}\mathbf{y}^T\right) = \boldsymbol{\mu}\boldsymbol{\mu}^T + \boldsymbol{\Sigma}_{\mathbf{y}}$$

$$* = Tr\left(\mathbf{B}\left(\boldsymbol{\mu}\boldsymbol{\mu}^T + \boldsymbol{\Sigma}_{\mathbf{y}}\right)\right) = Tr\left(\mathbf{B}\boldsymbol{\mu}\boldsymbol{\mu}^T\right) + Tr\left(\mathbf{B}\boldsymbol{\Sigma}_{\mathbf{y}}\right) =$$

$$= Tr\left(\boldsymbol{\mu}^T \mathbf{B} \boldsymbol{\mu}\right) + Tr\left(\mathbf{B}\boldsymbol{\Sigma}_{\mathbf{y}}\right) = \boldsymbol{\mu}^T \mathbf{B} \boldsymbol{\mu} + Tr\left(\mathbf{B}\boldsymbol{\Sigma}_{\mathbf{y}}\right)$$

Proof 2

$$\boxed{\text{if } \boldsymbol{y} \sim MN\left(\boldsymbol{\mu}, \boldsymbol{\Sigma}_{\boldsymbol{y}}\right) \rightarrow \left(\boldsymbol{y}-\boldsymbol{\mu}\right)^T \boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\left(\boldsymbol{y}-\boldsymbol{\mu}\right) \sim \chi^2\left(N\right)}$$

$$\left(\boldsymbol{y}-\boldsymbol{\mu}\right)^T \boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\left(\boldsymbol{y}-\boldsymbol{\mu}\right) = \left(\boldsymbol{y}-\boldsymbol{\mu}\right)^T \sum_{i=1,N} \frac{1}{\lambda_i} \boldsymbol{m}_i \boldsymbol{m}_i^T \left(\boldsymbol{y}-\boldsymbol{\mu}\right) = \sum_{i=1,N} \frac{\left(\boldsymbol{y}-\boldsymbol{\mu}\right)^T \boldsymbol{m}_i}{\sqrt{\lambda_i}} \frac{\boldsymbol{m}_i^T \left(\boldsymbol{y}-\boldsymbol{\mu}\right)}{\sqrt{\lambda_i}}$$

$$\frac{\boldsymbol{m}_i^T \left(\boldsymbol{y}-\boldsymbol{\mu}\right)}{\sqrt{\lambda_i}} = z_i \qquad \sum_{i=1,N} \frac{\left(\mathbf{y}-\boldsymbol{\mu}\right)^T \mathbf{m}_i}{\sqrt{\lambda_i}} \frac{\mathbf{m}_i^T \left(\mathbf{y}-\boldsymbol{\mu}\right)}{\sqrt{\lambda_i}} = \sum_{i=1,N} z_i^2$$

We know that $\sum_{i=1,N} z_i^2 \sim \chi^2\left(N\right)$ if $z_i \sim N\left(0,1\right)$ and $Cov\left(z_i, z_j\right) = 0$

z_i is Normal as it is a linear combination of normal variables

$$E\left(z_i\right) = \frac{\boldsymbol{m}_i^T E\left(\boldsymbol{y}-\boldsymbol{\mu}\right)}{\sqrt{\lambda_i}} = 0 \qquad V\left(z_i\right) = \frac{1}{\lambda_i} \boldsymbol{m}_i^T \boldsymbol{\Sigma}_{\boldsymbol{y}} \boldsymbol{m}_i = \frac{1}{\lambda_i} \boldsymbol{m}_i^T \lambda_i \boldsymbol{m}_i = 1$$

$$Cov\left(z_i, z_j\right) = \frac{\boldsymbol{m}_i^T E\left(\left(\boldsymbol{y}-\boldsymbol{\mu}\right)\left(\boldsymbol{y}-\boldsymbol{\mu}\right)^T\right) \boldsymbol{m}_j}{\sqrt{\lambda_i}\sqrt{\lambda_j}} = \frac{\boldsymbol{\Sigma}_{\boldsymbol{y}} \boldsymbol{m}_j}{\sqrt{\lambda_i}\sqrt{\lambda_j}} = \frac{\lambda_j}{\sqrt{\lambda_i}\sqrt{\lambda_j}} \boldsymbol{m}_i^T \boldsymbol{m}_j = 0$$

Confidence region for y $\quad \mathbf{y} \sim MN\left(\boldsymbol{\mu}, \boldsymbol{\Sigma}_y\right)$ and $\mathbf{y}$ is $N*1$

- We proved that $\quad \left(\mathbf{y}-\boldsymbol{\mu}\right)^T \boldsymbol{\Sigma}_y^{-1}\left(\mathbf{y}-\boldsymbol{\mu}\right) \sim \chi^2\left(N\right)$
- The **Confidence region** is $\quad \left(\mathbf{y}-\boldsymbol{\mu}\right)^T \boldsymbol{\Sigma}_y^{-1}\left(\mathbf{y}-\boldsymbol{\mu}\right) \leq \chi^2_\alpha\left(N\right)$
- What's the shape of this region?

$$\left(\mathbf{y}-\boldsymbol{\mu}\right)^T \boldsymbol{\Sigma}_y^{-1}\left(\mathbf{y}-\boldsymbol{\mu}\right) = \left(\mathbf{y}-\boldsymbol{\mu}\right)^T \sum_{i=1,N} \frac{1}{\lambda_i}\mathbf{m}_i\mathbf{m}_i^T\left(\mathbf{y}-\boldsymbol{\mu}\right) = \sum_{i=1,N} \frac{\left(\left(\mathbf{y}-\boldsymbol{\mu}\right)^T \mathbf{m}_i\right)^2}{\lambda_i}$$

- If $\quad t_i = \left(\mathbf{y}-\boldsymbol{\mu}\right)^T \mathbf{m}_i = \mathbf{m}_i^T\left(\mathbf{y}-\boldsymbol{\mu}\right)$
- The region boundary is an **ellipsoid** whose equation in the system of axes $m_1.m_2,\ m_N$ is:

$$\boxed{\sum_{i=1,N} \frac{t_i^2}{\lambda_i \chi^2_\alpha\left(N\right)} = 1}$$

The axes of the ellipsoid are the eigenvectors $\mathbf{m}_i$

Examples with N=2

- Example 1

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{11} \end{pmatrix} \rightarrow \begin{cases} \left|\mathbf{\Sigma} - \lambda \mathbf{I}\right| = 0 \rightarrow \left(\sigma_{11} - \lambda\right)^2 = 0 \rightarrow \lambda_{1,2} = \sigma_{11} \\ \mathbf{\Sigma m}_1 = \lambda_i \mathbf{m}_i \rightarrow \text{ we may use a couple of orthogonal axes} \\ \text{i.e. with versors } m_1 = \{0,1\} \text{ and } m_2 = \{1,0\} \end{cases}$$

- Example 2 NB: **Non normalized** eigenvectors

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix} \rightarrow \begin{cases} \lambda_{1,2} = \dfrac{1}{2}\left(\sigma_{11} + \sigma_{22} \pm \sqrt{\left(\sigma_{22} - \sigma_{11}\right)^2 + 4\sigma_{12}^2}\right) \\ \mathbf{m}_{1,2} = \begin{pmatrix} -\dfrac{-\sigma_{11} + \sigma_{22} \pm \sqrt{\left(\sigma_{22} - \sigma_{11}\right)^2 + 4\sigma_{12}^2}}{2\sigma_{12}} \\ 1 \end{pmatrix} \end{cases}$$

$$\mathbf{m}_{1,2} = \begin{Bmatrix} -\left(q \pm \sqrt{q^2 + 1}\right) \\ 1 \end{Bmatrix} \text{ with } q = \frac{\sigma_{22} - \sigma_{11}}{2\sigma_{12}}$$

- Case 1(it must be normalized) $\sigma_{11}=\sigma_{22}$

$$\begin{cases}\lambda_{1,2}=\dfrac{1}{2}\left(\sigma_{11}+\sigma_{22}\pm\sqrt{\left(\sigma_{22}-\sigma_{11}\right)^2+4\sigma_{12}^2}\right)\rightarrow\lambda_{1,2}=\sigma_{11}\pm\sigma_{12}\\ \mathbf{m}_{1,2}=\begin{pmatrix}-\dfrac{-\sigma_{11}+\sigma_{22}\pm\sqrt{\left(\sigma_{22}-\sigma_{11}\right)^2+4\sigma_{12}^2}}{2\sigma_{12}}\\ 1\end{pmatrix}=\begin{pmatrix}\mp1\\ 1\end{pmatrix}\end{cases}$$

- Case 2 (numerical)

$$\mathbf{\Sigma}=\begin{pmatrix}3 & 1\\ 1 & 4\end{pmatrix}\rightarrow\lambda_{1,2}=\frac{1}{2}\left(7\pm\sqrt{5}\right)=\begin{cases}4.618\\ 2.382\end{cases}$$

$$\mathbf{m}_{1,2}=\begin{pmatrix}\frac{1}{2}\left(-1\mp\sqrt{5}\right)\\ 1\end{pmatrix}\rightarrow\begin{cases}\mathbf{m}_1=\begin{pmatrix}-1.618\\ 1\end{pmatrix}\rightarrow\mathbf{m}_1^{Norm}=\begin{pmatrix}-0.851\\ 0.526\end{pmatrix}\\ \mathbf{m}_2=\begin{pmatrix}0.618\\ 1\end{pmatrix}\rightarrow\mathbf{m}_2^{Norm}=\begin{pmatrix}0.526\\ 0.851\end{pmatrix}\end{cases}$$

Regression Models

- **General Regression Model**

y *Response Variable* or «dependent variable»

d_j j^{th} **factor** $d_j \in \Re \rightarrow \mathbf{d} \in D^q \subset \Re^q$

Experimental Domain

x_i i^{th} **function** of **factors** or ***Regressor*** or "independent variable" or "**predictor**"

ε *Random term*

$\boldsymbol{\beta_i}$ *Regression Coefficients* (to be estimated)

Linear regression models

$$y = \beta_0 x_0\left(d_1, d_2, ..d_q\right) + \beta_1 x_1\left(d_1, d_2, ..d_q\right) + \ldots + \beta_k x_k\left(d_1, d_2, ..d_q\right) + \varepsilon$$

with $\quad \varepsilon \sim NID\left(0, \sigma^2\right)$

Examples

- A single Factor and polynomial models
- Two factors and 2^2 experiment

Linear model in vectorial form

$$\boxed{y = \mathbf{x}(\mathbf{d})^T \boldsymbol{\beta} + \varepsilon}$$

$$\mathbf{x} = \begin{Bmatrix} x_0(d_1, d_2, \ldots, d_q) \\ x_1(d_1, d_2, \ldots, d_q) \\ \ldots. \\ x_k(d_1, d_2, \ldots, d_q) \end{Bmatrix}$$

and

$$E(y) = \mathbf{x}^T \boldsymbol{\beta}$$

$$V(y) = \sigma^2$$

In the following: $\boldsymbol{p=k+1}$ -> Total number of parameters to be estimated

Examples of **linear models** (linear in parameters)

$$y = \beta_1 \ln d_1 + \beta_2 d_1^3 + \varepsilon \begin{cases} x_1 = \ln d_1 \\ x_2 = d_1^3 \end{cases}$$

$$z = \gamma_0 d_1^{\beta_1} e^{\varepsilon}$$

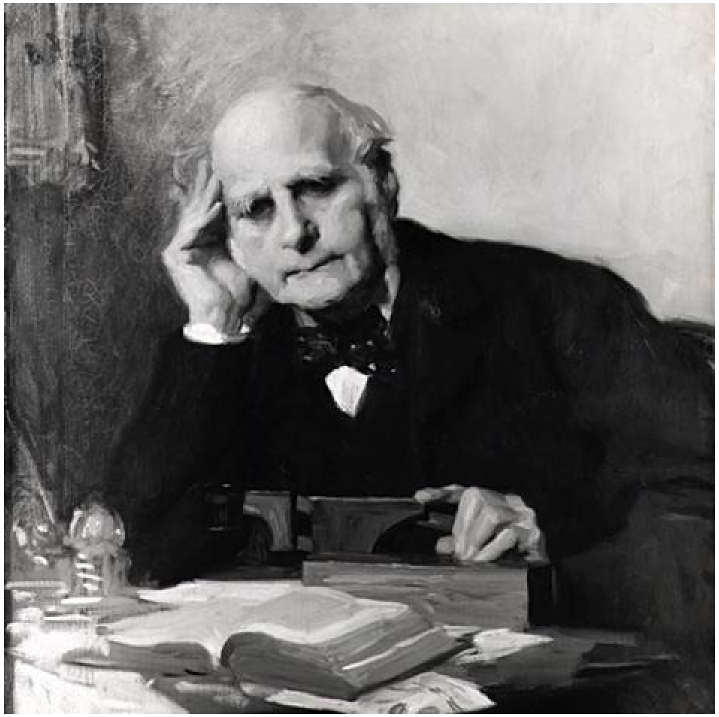
This is not a linear model

$$y = \beta_0 + \beta_1 d_1^{\beta_2} + \varepsilon$$

Regression

The word **regression** was firstly used by the english anthropologist **Sir Francis GALTON** in 1877

He published the work "**Typical Laws of Heredity**" in *Proceedings of the Royal Institution of Great Britain*, 8, pp. 282–301, (**1877**).



Francis Galton coined the term ***regression*** to describe situations in which there is reversion of a characteristic measured in offspring, ***away*** from the mean value of the *same* characteristic in ***their own*** parents, and ***towards*** the mean value in ***all*** offsprings.

He found an empirical relation between:

– $\boldsymbol{x}$ is the <u>parent</u> height average

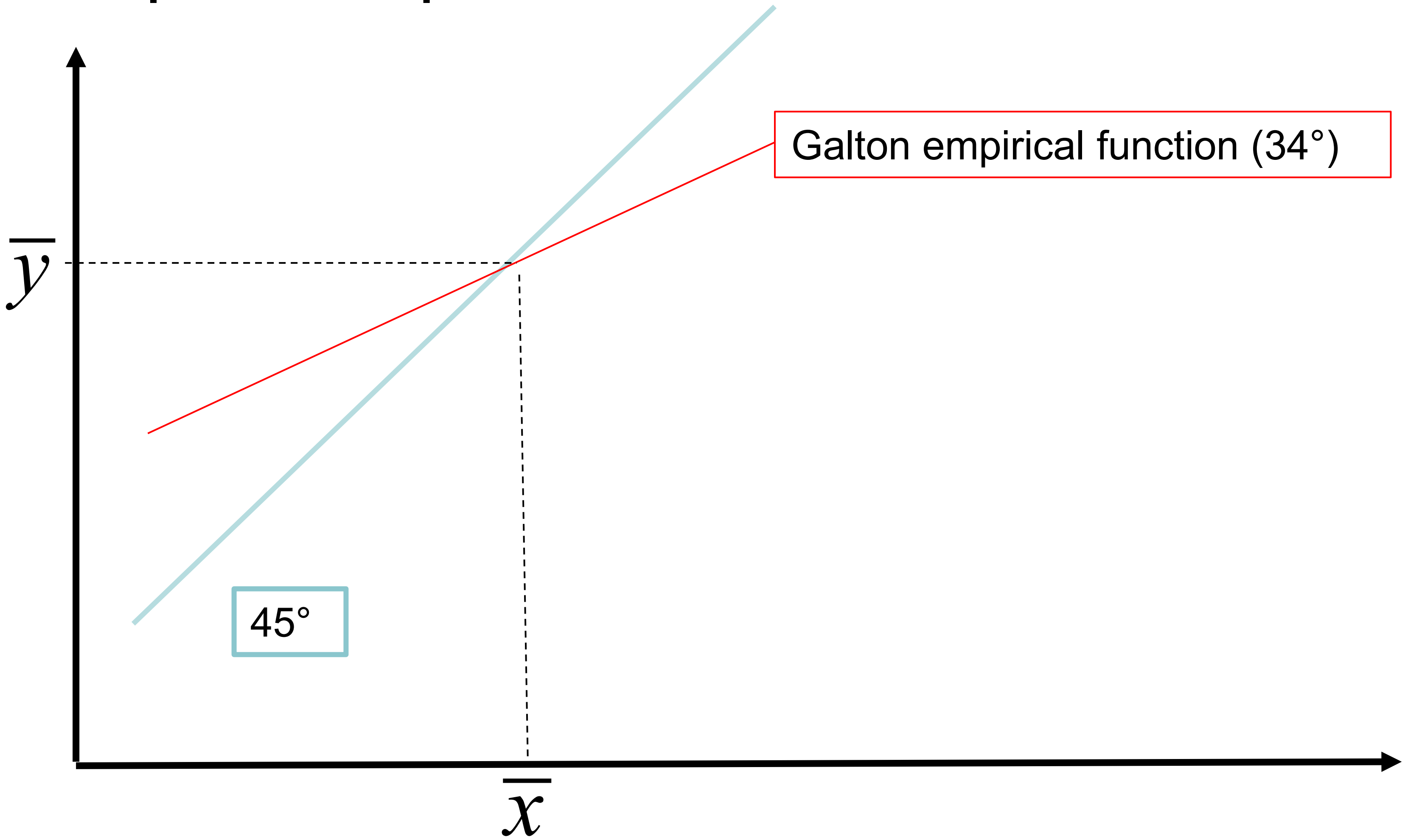
– $\boldsymbol{y}$ the <u>son</u> (offspring) height average.

$$y = \beta_0 + \beta_1 x + \varepsilon \quad \Leftarrow \quad \boxed{\hat{y} - \bar{y} = \frac{2}{3}\left(x - \bar{x}\right)}$$

$$\hat{\beta}_0 = \bar{y} - \frac{2}{3}\bar{x} \qquad \hat{\beta}_1 = \frac{2}{3}$$

$$\boxed{\hat{y} - \bar{y} = \frac{2}{3}\left(x - \bar{x}\right)}$$

- Graphical representation



(Training) Data set

- Treatment or Experimentation site with **q** Factors

$$\text{Treatment}_i = \left(d_{1i}, d_{2i}, \ldots, d_{qi} \right) = \mathbf{d}_i$$

- For each treatment i=1,$\boldsymbol{m}$ we could have $\boldsymbol{n_i}$ replicates

- For each treatment i we have n_i responses (observations)

$$\forall \mathbf{d}_i \rightarrow y_{ij} \qquad j = 1, n_i$$

- The Total number of observations is $N = \sum_{i=1,m} n_i$

Let us simplify the notation

$$\left.\begin{matrix}\mathbf{d}_i \\ \text{Model}\end{matrix}\right\} \rightarrow \mathbf{x}_i^T = \left(x_{0i}, x_{1i}, \ldots, x_{ki}\right)$$

Usually x_{0i}=1

$$\begin{pmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{31} \\ y_{32} \\ y_{33} \\ y_{34} \\ \cdots \\ y_{m2} \\ y_{m3} \end{pmatrix} \rightarrow \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \\ \cdots \\ y_{N-1} \\ y_N \end{pmatrix} \rightarrow \mathbf{y}$$

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{21} \ldots. + \beta_k x_{k1} + \varepsilon_1$$

$$y_2 = \beta_0 + \beta_1 x_{12} + \beta_2 x_{22} \ldots. + \beta_k x_{k2} + \varepsilon_2$$

....

$$y_N = \beta_0 + \beta_1 x_{1N} + \beta_2 x_{2N} \ldots. + \beta_k x_{kN} + \varepsilon_N$$

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i \text{ or } y_i = \boldsymbol{\beta}^T \mathbf{x}_i + \varepsilon_i \text{ with } i = 1, N$$

Methods for Estimating model parameters

- Maximum Likelihood method
- OLS Ordinary Least Squares
- WLS Weighted Least Squares
- LOESS (LOcally Estimated Scatterplot Smoothing)
- LOWESS (LOcally WEighted Scatterplot Smoothing)
- ….

Maximum Likelihood method

Let us consider a discrete random variable whose distribution is:

Yi	4	5	6	7	8
$Pr(Y_i)$	1/20	1/20	5/20	5/20	8/20

Let us draw two samples of 6 independent numbers:

S1: {4, 4, 5, 6, 5, 5} and S2: {6, 7, 8, 4, 7, 8)

Compute the probability of each sample

$Pr(S1)=(1/20)^5 \times (5/20)=7.81\times10^{-8}$

$Pr(S2)=(5/20)^3 \times (8/20)^2 \times(1/20)=1.25\times10^{-4}$

The two formulas compute the **sample joint distribution** of the observed samples

Example: Binomial Distribution

We can use the **sample joint distribution** to estimate the probability distribution parameters

Let us consider a **binomial distribution**:

$$B\left(b \mid n, p\right) = \binom{n}{b} p^{b} \left(1 - p\right)^{n-b}$$

we want to estimate the parameter **p** from a sample of N independent drawings $(b_1, b_2, \ldots b_N)$

The **sample joint (probability) distribution** is:

$$L\left(b_i, b_2, .., b_N \mid n, p\right) = B\left(b_1 \mid n, p\right) * .. * B\left(b_N \mid n, p\right) =$$

$$= \binom{n}{b_1}\binom{n}{b_2} .. \binom{n}{b_n} p^{\sum_{i=1,N} b_i} \left(1 - p\right)^{Nn - \sum_{i=1,N} b_i} = A\left(\mathbf{b}\right) p^{b.} \left(1 - p\right)^{Nn - b.}$$

We want estimate ***p*** by solving the following problem:

$$\hat{p} = \arg \underset{p}{Max}\, L(\mathbf{b} \mid n, p)$$

Or a monotonic transformation of L->l

$$l(\mathbf{b} \mid n, p) = \ln(\mathbf{b} \mid n, p) \rightarrow \hat{p} = \arg \underset{p}{Max}\, l(\mathbf{b} \mid n, p)$$

In our case:

$$l(\mathbf{b} \mid n, p) = \ln A(\mathbf{b}) + b_{\bullet} \ln p + (Nn - b_{\bullet}) \ln(1 - p)$$

$$\frac{\partial l(\mathbf{b} \mid n, p)}{\partial p} = \frac{b_{\bullet}}{p} - (Nn - b_{\bullet}) \frac{1}{1 - p} = 0$$

$$b_{\bullet}(1 - p) - p(Nn - b_{\bullet}) = 0 \rightarrow \hat{p} = \frac{b_{\bullet}}{Nn}$$

Continuous Random Variable

Let us suppose that Y follows a **density function**

$$Y \sim f\left(y \,|\, \boldsymbol{\theta}\right) \ \textbf{where} \ \begin{cases} \boldsymbol{\theta} & \text{parameters to be estimated} \\ f & \text{probability density function} \end{cases}$$

1) Build the **Sample likelihood** function

$$L(\boldsymbol{\theta}) = \prod_{i=1,N} f(y_i \,|\, \boldsymbol{\theta})$$

Note: We loose the meaning of sample probability

2) Maximize the likelihood function or a monotonic transformation of it (the so called LogLikelihood)

3) Verify the estimator unbiasedness and change if necessary

Example: Model: $y_i = \boldsymbol{\beta}^T \mathbf{x}_i + \varepsilon_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i$

Normal density assumption

$$f\left(y_i \mid \boldsymbol{\beta}, \sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{y_i - E\left(y_i\right)}{\sigma}\right)^2} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{y_i - \mathbf{x}_i^T \boldsymbol{\beta}}{\sigma}\right)^2}$$

Let us build the **Likelihood function**

$$L(\boldsymbol{\beta}, \sigma^2) = \prod_{i=1,N} f(y_i \mid \boldsymbol{\beta}, \sigma^2) = \frac{1}{\left(2\pi\right)^{N/2} \left(\sigma^2\right)^{N/2}} e^{-\frac{1}{2\sigma^2} \sum_{i=1,N} \left(y_i - \mathbf{x}_i^T \boldsymbol{\beta}\right)^2}$$

Transform **(monotonic transformation)**

$$l\left(\boldsymbol{\beta},\sigma^2\right)=\ln\left[L(\boldsymbol{\beta},\sigma^2)\right]=-\frac{N}{2}\ln\left(2\pi\right)-\frac{N}{2}\ln\left(\sigma^2\right)-\frac{1}{2\sigma^2}\sum_{i=1,N}\left(y_i-\boldsymbol{x}_i^T\boldsymbol{\beta}\right)^2$$

Find the **maximum** through the **stationary conditions**

$$\frac{\partial l\left(\boldsymbol{\beta},\sigma^2\right)}{\partial\sigma^2}=-\frac{N}{2\hat{\sigma}^2}+\frac{\sum_{i=1,N}\left(y_i-\boldsymbol{x}_i^T\hat{\boldsymbol{\beta}}\right)^2}{2\hat{\sigma}^4}=0\Rightarrow\Rightarrow\hat{\sigma}^2=\frac{\sum_{i=1,N}\left(y_i-\boldsymbol{x}_i^T\hat{\boldsymbol{\beta}}\right)^2}{N}$$

$$\frac{\partial l\left(\boldsymbol{\beta},\sigma^2\right)}{\partial\boldsymbol{\beta}}=0\Rightarrow\Rightarrow$$ the solution minimizes $\sum_{i=1,N}\left(y_i-\boldsymbol{x}_i^T\boldsymbol{\beta}\right)^2$

⇧

Ordinary Least Square OLS

Ordinary Least Squares (OLS)

- To **estimate the model parameters** we use OLS (equivalent to Maximum likelihood since we are assuming that the observations are **normally distributed**)
- Build the quantity

$$Q(\boldsymbol{\beta}) = \sum_{i=1,N} \varepsilon_i^2 = \sum_{i=1,N} \left(y_i - E(y_i) \right)^2 = \sum_{i=1,N} \left(y_i - \mathbf{x}_i^T \boldsymbol{\beta} \right)^2$$

If we assume that x_{i0}=1

$$\text{then} \quad \mathbf{x}_i^T \boldsymbol{\beta} = \beta_0 + x_{i1}\beta_1 + x_{i2}\beta_2 + \ldots x_{ik}\beta_k$$

Then we minimize $Q(\boldsymbol{\beta})$ and we obtain the $\boldsymbol{\beta}$ estimators

Note: the number of the model parameters is $\boldsymbol{p=k+1}$

Let us derive

$$Q=\sum_i \varepsilon_i^2=\sum_{i=1}^{N}(y_i-\beta_0-\sum_{j=1}^{k}\beta_j x_{ij})^2$$

$$\frac{\partial Q}{\partial \beta_0}=-2\sum_{i=1}^{N}\left(y_i-\hat{\beta}_0-\sum_{j=1}^{k}\hat{\beta}_j x_{ij}\right)=0$$

....

$$\frac{\partial Q}{\partial \beta_j}=-2\sum_{i=1}^{N}x_{ij}\left(y_i-\hat{\beta}_0-\sum_{j=1}^{k}\hat{\beta}_j x_{ij}\right)=0$$

(After some algebric manipulation) **Normal Equations**

$$\sum_i y_i = \mathrm{N}\hat{\beta}_0 + \hat{\beta}_1\sum_i x_{i1} + \hat{\beta}_2\sum_i x_{i2} + \ldots\ldots + \hat{\beta}_k\sum_i x_{ik}$$

$$\sum_i x_{i1}y_i = \hat{\beta}_0\sum_i x_{i1} + \hat{\beta}_1\sum_i x_{i1}^2 + \hat{\beta}_2\sum_i x_{i1}x_{i2} + \ldots\ldots + \hat{\beta}_k\sum_i x_{i1}x_{ik}$$

....

$$\sum_i x_{ik}y_i = \hat{\beta}_0\sum_i x_{ik} + \hat{\beta}_1\sum_i x_{ik}x_{i1} + \hat{\beta}_2\sum_i x_{ik}x_{i2} + \ldots\ldots\hat{\beta}_k\sum_i x_{ik}^2$$

Matrix Approach

Let us start with:

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i$$

$$\mathbf{y} = \begin{Bmatrix} y_1 \\ y_2 \\ . \\ y_N \end{Bmatrix} \quad \boldsymbol{\beta} = \begin{Bmatrix} \beta_0 \\ \beta_1 \\ . \\ \beta_k \end{Bmatrix} \quad \mathbf{X} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ . \\ \mathbf{x}_N^T \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & . & x_{1k} \\ 1 & x_{21} & . & x_{2k} \\ . & . & . & . \\ 1 & x_{N1} & . & x_{Nk} \end{bmatrix} \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ . \\ \varepsilon_N \end{bmatrix}$$

We can represent the **training data set** and the model In a more compact form

$\mathbf{X}$ is called **Model Matrix**

$$\boxed{\begin{array}{l} \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad \text{with} \quad \mathbf{V}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I} \;\; E(\boldsymbol{\varepsilon}) = \mathbf{0} \\ \text{Note that} \rightarrow E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta} \;\; V(\mathbf{y}) = \boldsymbol{\Sigma}_{\mathbf{y}} = \sigma^2\mathbf{I} \end{array}}$$

Let us minimize $\mathbf{Q(\beta)}$ (squared lenght of the $\boldsymbol{\varepsilon}$ vector)

$$Q(\boldsymbol{\beta}) = \sum_i \varepsilon_i^2 = \boldsymbol{\varepsilon}^T\boldsymbol{\varepsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = (\mathbf{y}^T - \boldsymbol{\beta}^T\mathbf{X}^T)(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) =$$

$$= y^T y - y^T X\beta - \beta^T X^T y + \beta^T X^T X\beta$$

$$= y^T y - 2\beta^T X^T y + \beta^T X^T X\beta$$

Then derive and pose the derivative equal to zero

$$\frac{\partial Q}{\partial \boldsymbol{\beta}} = 0 \Rightarrow -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = 0$$

$$\boxed{\left(\mathbf{X}^T\mathbf{X}\right)\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}} \Rightarrow \textbf{NORMAL EQUATIONS}$$

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}}$$

X^TX features

- $\mathbf{X^TX}$ Matrix dimension is (p x p) with p=k+1
- It's a symmetrical matrix
- If the rank of $\mathbf{X}$ is **p** (full column rank), $(\mathbf{X^TX})$ can be inverted and its inverse is symmetrical.
- Note: full column rank **when each of the columns of the matrix X are linearly <u>independent</u> of the other columns**

- **Row representation**

$$\mathbf{X}^T\mathbf{X} = \sum_{i=1,N} \mathbf{x}_i \mathbf{x}_i^T$$

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \dots \\ \mathbf{x}_N^T \end{bmatrix} \quad \mathbf{X}^T = \begin{bmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_N \end{bmatrix}$$

- **Column representation** $\mathbf{X} = \begin{bmatrix} \mathbf{X}_0 & \mathbf{X}_1 & \dots & \mathbf{X}_k \end{bmatrix}$

$$\mathbf{X}^T = \begin{bmatrix} \mathbf{X}_0^T \\ \mathbf{X}_1^T \\ \dots \\ \mathbf{X}_k^T \end{bmatrix} \qquad \mathbf{X}^T\mathbf{X} = \begin{bmatrix} \mathbf{X}_0^T\mathbf{X}_0 & \mathbf{X}_0^T\mathbf{X}_1 & \dots & \mathbf{X}_0^T\mathbf{X}_k \\ \mathbf{X}_1^T\mathbf{X}_0 & \mathbf{X}_1^T\mathbf{X}_1 & \dots & \mathbf{X}_1^T\mathbf{X}_k \\ \dots & \dots & \dots & \dots \\ \mathbf{X}_k^T\mathbf{X}_0 & \mathbf{X}_k^T\mathbf{X}_1 & \dots & \mathbf{X}_k^T\mathbf{X}_k \end{bmatrix}$$

- If the Columns of $\mathbf{X}$ are orthogonal the matrix $\mathbf{X}^\mathrm{T}\mathbf{X}$ is diagonal
- **Example:** replicated 2^2 experiment with a full model

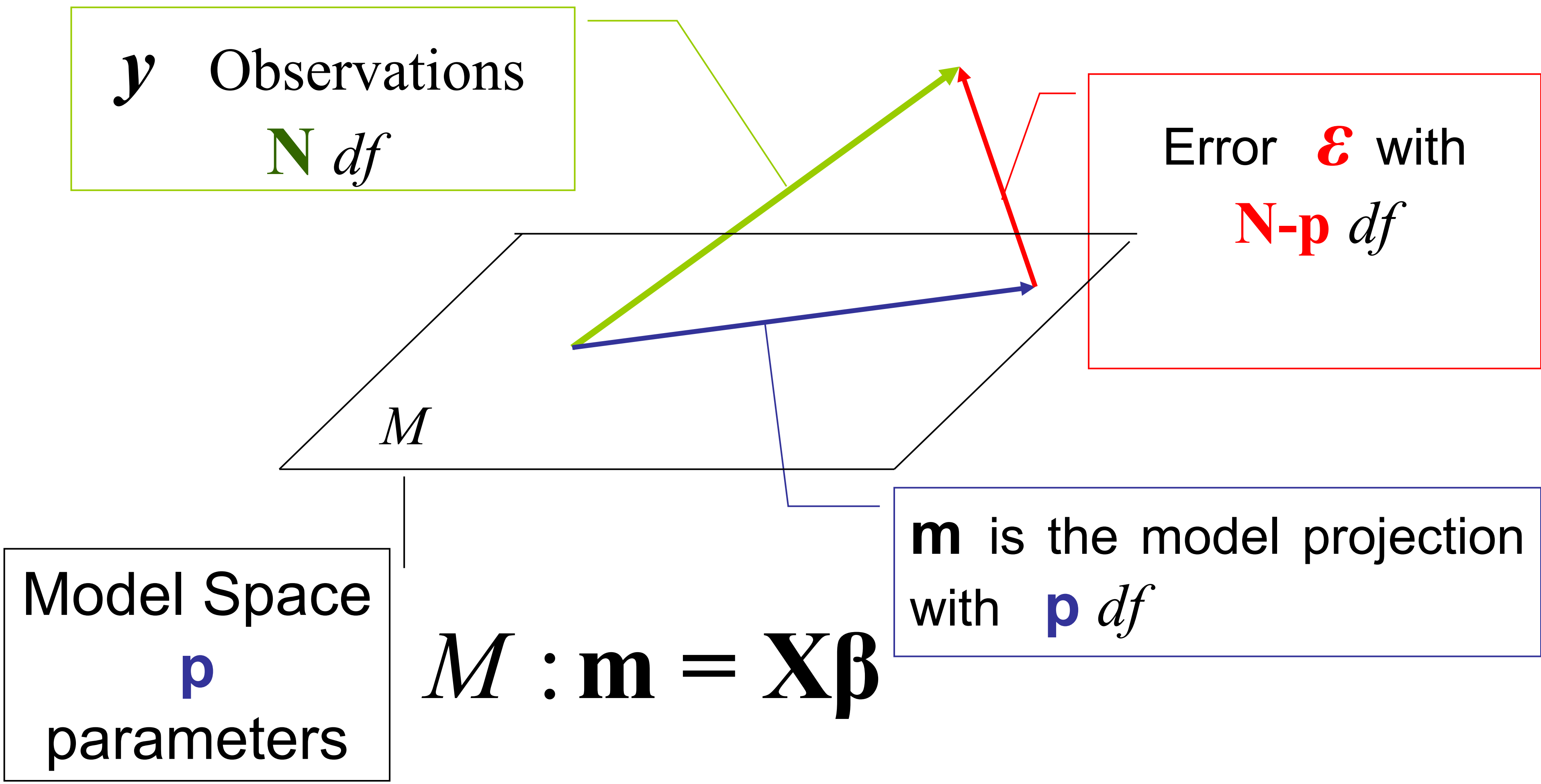
Properties of $\boxed{\hat{\boldsymbol{\beta}} = \left(\boldsymbol{X}^T \boldsymbol{X}\right)^{-1} \boldsymbol{X}^T \boldsymbol{y}}$

- The vector is a **linear combinations** of the random observations $\boldsymbol{y}$ so it is a **random vector**
- If the observations are normal, the vector is **Multivariate Normal distributed**

$$\boxed{\begin{aligned} E(\hat{\boldsymbol{\beta}}) &= \boldsymbol{\beta} \\ V(\hat{\boldsymbol{\beta}}) &= \Sigma_{\hat{\beta}} = (X^T X)^{-1} \sigma^2 \end{aligned}}$$

$$\boldsymbol{\varepsilon} \sim MN(0, \sigma^2 \mathbf{I}) \Longrightarrow \hat{\boldsymbol{\beta}} \sim MN(\boldsymbol{\beta}, (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1} \sigma^2)$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \Rightarrow \mathbf{y} = \mathbf{X_0}\beta_0 + \mathbf{X_1}\beta_1 + \ldots + \mathbf{X_k}\beta_k + \boldsymbol{\varepsilon}$$



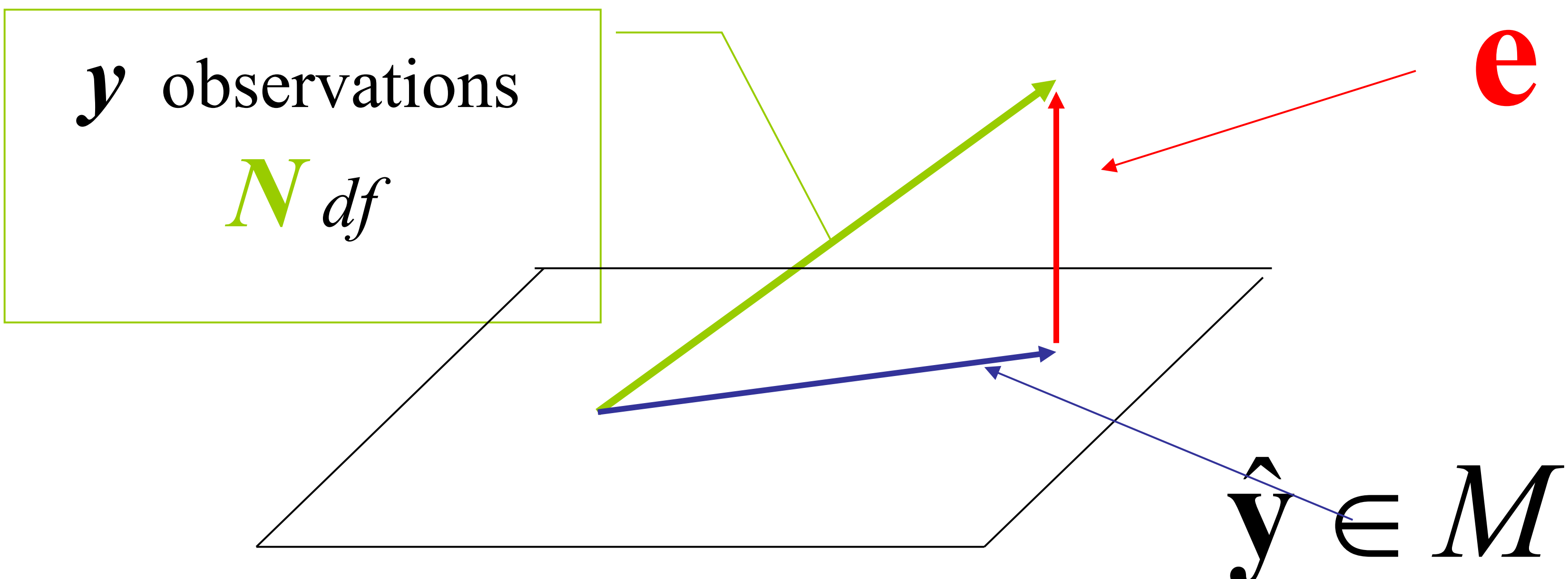
$$M : \mathbf{m} = \mathbf{X}\boldsymbol{\beta}$$

How to choose **m** (projection of y onto M)?

Orthogonal projection onto the model space X

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{y} = \mathbf{P}_{\mathbf{M}}\mathbf{y}$$

$$\mathbf{P}_{\mathbf{M}} = \mathbf{X}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\mathbf{X}^{\mathrm{T}} \quad \text{Projection Matrix onto M}$$



Projection matrix properties

$\mathbf{P_M} = \mathbf{P_M^T} \rightarrow$ Symmetric with NxN dimensions

$\mathbf{P_M^T P_M} = \mathbf{P_M^2} = \mathbf{P_M} \rightarrow$ Idempotent

$\mathbf{P_M} \equiv \mathbf{H} = \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T$ Hat Matrix $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$

$$\boxed{h_{ij} = \mathbf{x}_i^T\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{x}_j}$$

if there is a constant in the model

$\mathbf{H1} = \mathbf{1} \quad \mathbf{1}^T\mathbf{H} = \mathbf{1}^T \quad \mathbf{1}^T\mathbf{H1} = N$

Proof: **H1=1**

$\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{X} = \mathbf{X}$ *or* $\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\left[\mathbf{X}_0, \mathbf{X}_1, \ldots, \mathbf{X}_k\right] = \left[\mathbf{X}_0, \mathbf{X}_1, \ldots, \mathbf{X}_k\right]$

$\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{X}_0 = \mathbf{X}_0$ *where* $\mathbf{X}_0 = \mathbf{1}$ *if there is a constant in the model*

Remark: we can project a matrix by projecting each column

- We can also have a projection matrix onto the **Error space** (orthogonal to M)

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y} \rightarrow \mathbf{P_E} = \mathbf{I} - \mathbf{H}$$

$$\boxed{\mathbf{e} \text{ and } \hat{\mathbf{y}} \text{ are orthogonal vectors } \rightarrow \mathbf{e}^{\mathbf{T}}\hat{\mathbf{y}} = 0}$$

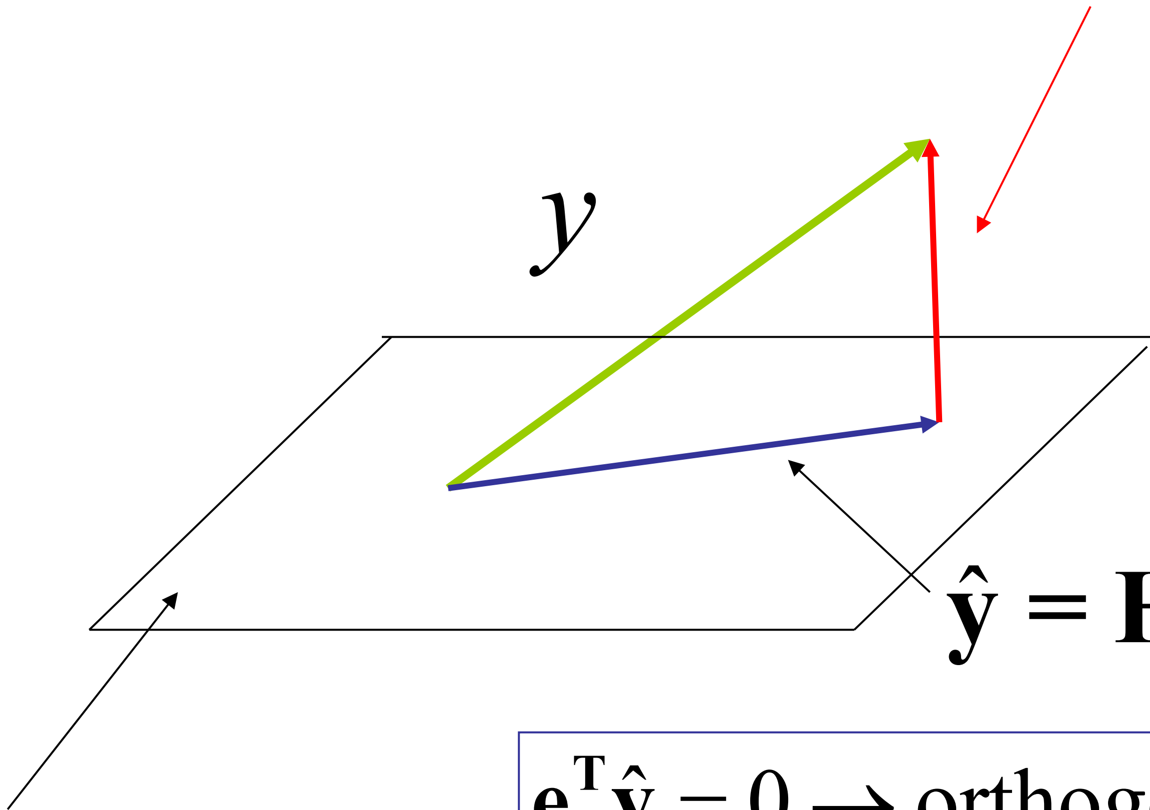
- The two **projection matrices** are orthogonals:

$$\mathbf{P_E^T P_M} = (\mathbf{I} - \mathbf{H})^{\mathbf{T}}\mathbf{H} = \mathbf{IH} - \mathbf{H^T H} = \mathbf{H} - \mathbf{H} = \mathbf{0}$$

- We can easily prove:

$$\left.\begin{aligned} E(\mathbf{e}) &= \mathbf{0} \\ \mathbf{V}(\mathbf{e}) = \mathbf{\Sigma_e} &= (\mathbf{I} - \mathbf{H})\sigma^2 \end{aligned}\right\} \Rightarrow \quad \rho_{ij} = \frac{-h_{ij}}{\sqrt{(1-h_{ii})(1-h_{jj})}}$$

$$\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$$



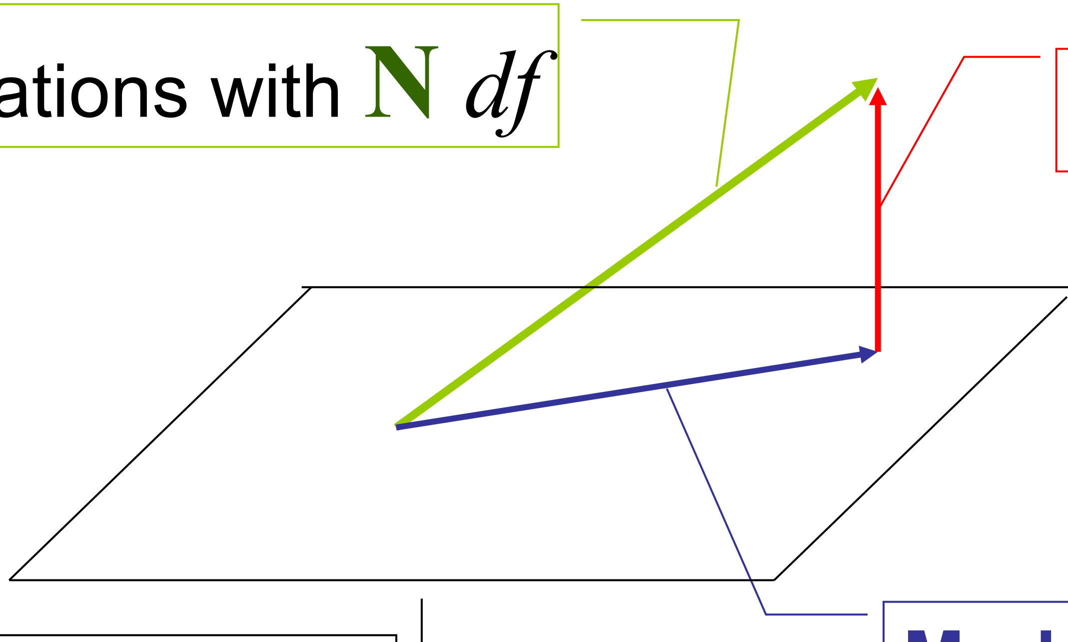
$$y$$

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$$

$$\boxed{\mathbf{e}^{\mathrm{T}}\hat{\mathbf{y}} = 0 \rightarrow \text{orthogonal vectors}}$$

$$M : \mathbf{m} = \mathbf{X}\boldsymbol{\beta}$$

Pythagoras' theorem



$$\mathbf{y}^T\mathbf{y} = \hat{\mathbf{y}}^T\hat{\mathbf{y}} + \mathbf{e}^T\mathbf{e}$$

$$SS^*_{TOT} = SS^*_{REG} + SS_E$$

- **Pythagoras' theorem elements**

$$SS^*_{TOT} = \mathbf{y}^{\mathrm{T}}\mathbf{y} \quad \text{with} \quad df^*_{TOT} = N$$

$$SS^*_{REG} = \begin{cases} \hat{\mathbf{y}}^T\hat{\mathbf{y}} \\ \mathbf{y}^T\mathbf{H}\mathbf{y} \\ \hat{\boldsymbol{\beta}}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}_{\hat{\boldsymbol{\beta}}}\hat{\boldsymbol{\beta}}\sigma^2 \\ \hat{\boldsymbol{\beta}}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{y} \end{cases} \quad \text{with} \quad df^*_{REG} = p$$

$$SS_E = \mathbf{e}^T\mathbf{e} = \begin{cases} \mathbf{y}^T\mathbf{y} - SS^*_{REG} \\ \mathbf{y}^T\left(\mathbf{I} - \mathbf{H}\right)\mathbf{y} \end{cases} \quad \text{with} \quad df_E = N - p$$

Variance estimator

- Variance estimator:

$$\hat{\sigma}^2 = \frac{SS_E}{df_E} = \frac{\mathbf{e}^T\mathbf{e}}{N-p} = \frac{\mathbf{y}^T\left(\mathbf{I}-\mathbf{H}\right)\mathbf{y}}{N-p}$$

- Let us see if it is unbiased

$$E\left(\hat{\sigma}^2\right) = \frac{E\left(\mathbf{y}^T\left(\mathbf{I}-\mathbf{H}\right)\mathbf{y}\right)}{N-p} = \frac{1}{N-p}\left[\boldsymbol{\beta}^T\mathbf{X}^T\left(\mathbf{I}-\mathbf{H}\right)\mathbf{X}\boldsymbol{\beta} + Tr\left(\left(\mathbf{I}-\mathbf{H}\right)\boldsymbol{\Sigma}_y\right)\right] = \sigma^2$$

Studentized residuals

$$\boxed{\mathbf{StdRes}_i = \frac{e_i}{\sqrt{\hat{V}(e_i)}}}$$

$$\boxed{V(\mathbf{e}) \equiv \boldsymbol{\Sigma}_{\mathbf{e}} = (\mathbf{I} - \mathbf{H})\sigma^2}$$

$$\hat{V}(e_i) = (1 - h_{ii})\hat{\sigma}^2$$

$$\boxed{\mathrm{StdRes}_i = r_i = \frac{e_i}{\sqrt{\hat{\sigma}^2 (1 - h_{ii})}}}$$

a) Studentized residuals are correlated

b) Diagonal terms are usually **not equals**: different error variances

c) Usual rule on the thresholds of studentized residuals (± 3)

d) They are called **Internally studentized residuals**

Residual checking

- Check if **StdRes** have values outside the usual interval [-3,3]
- Plot **StdRes** vs Fits and Predictor values
- **Normality** test (minimum 50 data, for good power)
- **Variance homogeneity** test if we have replicates (Bartlett, Brown-Forsyte).
- **Autocorrelation** test (Bartlett, Ljung-Box, Durbin-Watson, minimum 50 data) if you know the run order

- Some disputes this approach. Why?
- Residuals are computed using all data

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$$

$$e_i = (1 - h_{ii}) y_i + \sum_{\substack{l=1,N \\ l \neq i}} (-h_{il} y_l)$$

$$e_i = f_i(y_1, y_2, \ldots, \color{red}{y_i}\color{black}, \ldots, y_N)$$

- The idea is to compute the $\boldsymbol{i_{th}}$ residuals **eliminating the $\boldsymbol{i_{th}}$** observation

$$e_{(i)} = f_{(i)}(y_1, y_2, \ldots, y_{i-1}, y_{i+1}, \ldots, y_N)$$

PRESS (PRediction Error Sum of Squares)

Take **N-1** observations out of **N**

Estimate the regression model

Predict the **left out** data

Compute the residual for the left out data

$$e_{(i)} = y_i - \hat{y}_{i(i)}$$

i=1,N

$$PRESS = \sum_{i=1}^{N} e_{(i)}^2$$

Instead of calculating N regressions, someone proved (we do not do it) that:

$$e_{(i)} = \frac{e_i}{1-h_{ii}} \quad \rightarrow PRESS = \sum_{1=1,N} e_{(i)}^2 = \sum_{i=1,N} \left(\frac{e_i}{1-h_{ii}} \right)^2$$

We can define an index to evaluate the goodness of the model **predictive capability**

$$\Rightarrow \quad R_{PRED}^2 = 1 - \frac{PRESS}{SS_{TOT}}$$

Cross-Validation

- What we have just discussed is a particular technique of **Cross Validation** named **leave-one-out**
- Several procedures leaving **k observations at a time** have been proposed.
- Remark: if you have replicates, you must remove (each time) **all replications** of the same experimental condition (treatment).

External Studentized Residuals

$s_{(i)}^2$ estimates σ^2 leaving out the i_{th} observation

$$s_{(i)}^2 = \frac{SS_E - SS(e_i)}{(N-p)-1} = \frac{(N-p)MS_E - e_i^2/(1-h_{ii})}{N-p-1}$$

The **External Studentized Residual** is

$$t_i = \frac{e_i}{\sqrt{s_{(i)}^2(1-h_{ii})}} \sim t(N-p-1)$$

MTB call them **TRES**

RESIDUALS	INTERNAL prediction model	EXTERNAL prediction model
NOT STUDENTIZED	$e_i = y_i - \hat{y}_i$ $SS_E = \sum_{i=1,N} e_i^2$ $MS_E = \frac{SS_E}{(N-p)}$	$e_{(i)} = y_i - \hat{y}_{(i)} = \frac{e_i}{1-h_{ii}}$ $PRESS = \sum_{i=1,N} e_{(i)}^2$
STUDENTIZED	**SRES** $r_i = \frac{e_i}{\sqrt{MS_E(1-h_{ii})}}$	**TRES** $t_i = \frac{e_i}{\sqrt{s_{(i)}^2(1-h_{ii})}}$ $s_{(i)}^2 = \frac{(N-p)MS_E - e_i^2/(1-h_{ii})}{N-p-1}$

How to choose a good model?

A. Statistical **significance** of the «**«Regression Model»**

B. How to **improve** the **Model** (<u>expanding</u> or <u>reducing</u>)

C. Goodness of the Model **prediction** capability

Regression Model Significance

Hypotheses

$$H_0 : \ \boldsymbol{\beta} = \mathbf{0}$$

$$H_1 : \ \boldsymbol{\beta} \neq \mathbf{0}$$

Sum of Squares partition is

$$SS^*_{TOT} = SS^*_{REG} + SS_E$$

$$MS^*_{REG} = \frac{SS^*_{REG}}{df^*_{REG}} \qquad \hat{\sigma}^2 = MS_E = \frac{SS_E}{df_E}$$

$$\boxed{\text{if } H_0 \text{ holds} \quad \Rightarrow \quad \frac{MS^*_{REG}}{MS_E} \sim F(p, N-p)}$$

Proof

- Pythagoras' theorem $SS_{TOT}^{*} = SS_{REG}^{*} + SS_{E}$
- Orthogonal segments $\Rightarrow$ SS_{REG}^{*} SS_{E} independents

$$SS_{TOT}^{*} = \mathbf{y}^T\mathbf{y} = \mathbf{y}^T\boldsymbol{\Sigma}_{\mathbf{y}}^{-1}\mathbf{y}\sigma^2 \quad \text{if } \mathbf{y} \sim N\left(\mathbf{X}\boldsymbol{\beta} = 0, \mathbf{I}\sigma^2\right) \rightarrow \frac{SS_{TOT}^{*}}{\sigma^2} \sim \chi^2(N)$$

$$SS_{REG}^{*} = \hat{\mathbf{y}}^T\hat{\mathbf{y}} = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^T\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}}^{-1}\hat{\boldsymbol{\beta}}\sigma^2$$

$$\frac{SS_{REG}^{*}}{\sigma^2} = \hat{\boldsymbol{\beta}}^T\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}}^{-1}\hat{\boldsymbol{\beta}} \sim \chi^2(p) \text{ if } \hat{\boldsymbol{\beta}} \sim N\left(\mathbf{0}, \boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}}\right)$$

$$\boxed{\frac{SS_E}{\sigma^2} = \frac{SS_{TOT}^{*}}{\sigma^2} - \frac{SS_{REG}^{*}}{\sigma^2} = \chi^2(N) - \chi^2(p) \sim \chi^2(N-p)}$$

$$\frac{\dfrac{1}{p}\dfrac{SS_{REG}^{*}}{\sigma^2}}{\dfrac{1}{N-p}\dfrac{SS_E}{\sigma^2}} = \frac{MS_{REG}^{*}/\sigma^2}{MS_E/\sigma^2} = \frac{MS_{REG}^{*}}{MS_E} = \frac{\dfrac{\chi^2(p)}{p}}{\dfrac{\chi^2(N-p)}{(N-p)}} \sim F(p, N-p)$$

Contribution of the Average

- Consider the simple model $y = \beta_0 + \varepsilon$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{1}\beta_0 + \boldsymbol{\varepsilon}$$

$$\hat{\beta}_0 = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{y} = \left(\mathbf{1}^T\mathbf{1}\right)^{-1}\mathbf{1}^T\mathbf{y} = y_{\bullet} / N = \overline{y}_{\bullet}$$

$$SS^*_{REG} = \hat{\boldsymbol{\beta}}^T\left(\mathbf{1}^T\mathbf{1}\right)\hat{\boldsymbol{\beta}} = \frac{y_{\bullet}}{N} N \frac{y_{\bullet}}{N} = \frac{y_{\bullet}^2}{N}$$

- Consider the SS_{TOT} of data, **corrected** with the mean

$$SS_{TOT} = \left(\mathbf{y} - \frac{y_{\bullet}}{N}\mathbf{1}\right)^T\left(\mathbf{y} - \frac{y_{\bullet}}{N}\mathbf{1}\right)$$

$$= \mathbf{y}^T\mathbf{y} - \frac{y_{\bullet}}{N}\mathbf{y}^T\mathbf{1} - \frac{y_{\bullet}}{N}\mathbf{1}^T\mathbf{y} + \left(\frac{y_{\bullet}}{N}\right)^2\mathbf{1}^T\mathbf{1} = \mathbf{y}^T\mathbf{y} - \frac{y_{\bullet}^2}{N}$$

Usually **we exclude** the model constant β_0 from the Null Hypothesis

$$H_0: \quad \beta_1 = \beta_2 = ... = \beta_k = 0$$

$$H_1: \quad \beta_j \neq 0 \quad \text{for at least one } j = 1, 2, ...k$$

$$SS_{TOT} = \mathbf{y}^{\mathrm{T}}\mathbf{y} - \frac{y_{\bullet}^2}{N} \qquad df_{TOT} = N - 1$$

$$SS_{REG} = \hat{\boldsymbol{\beta}}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{y} - \frac{y_{\bullet}^2}{N} \qquad df_{REG} = p - 1 = k$$

$$SS_E = \mathbf{y}^{\mathrm{T}}\mathbf{y} - \hat{\boldsymbol{\beta}}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{y} \qquad df_E = N - p$$

$$MS_{REG} = \frac{SS_{REG}}{df_{REG}} \qquad \hat{\sigma}^2 = MS_E = \frac{SS_E}{df_E}$$

$$\boxed{H_0 \text{ holds} \rightarrow \frac{MS_{REG}}{MS_E} \sim F(p-1, N-p)}$$

ANOVA Table

Source	SS	*df*	MS	F_0
Regression (corrected)	SS_{REG} $\quad \hat{\boldsymbol{\beta}}^T \mathbf{X}^T \mathbf{y} - \frac{y_{\bullet}^2}{N}$	p-1	MS_{REG}	F_0
Error	SS_E $\quad \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T \mathbf{X}^T \mathbf{y}$	N-p	MS_E	
Total (corrected)	SS_{TOT} $\quad \mathbf{y}^T\mathbf{y} - \frac{y_{\bullet}^2}{N}$	N-1		

$$MS_{REG} = \frac{SS_{REG}}{p-1} \qquad MS_E = \frac{SS_E}{N-p} \qquad F_o = \frac{MS_{REG}}{MS_E}$$

Heuristic rule
(G.E.P. Box)

$$\Rightarrow \quad F_0 > 4F_\alpha\left(p-1, N-p\right)$$

R^2 e R^2_{adj} Coefficients

Quite often people use **R²** (**Multiple Correlation Coefficient or Coefficient of determination**)

$$R^2 = \frac{SS_{REG}}{SS_{TOT}} = 1 - \frac{SS_E}{SS_{TOT}}$$

A better choice is R^2_{adj}

$$R^2_{adj} = 1 - \frac{SS_E/(N-p)}{SS_{TOT}/(N-1)} = 1 - \left(1 - R^2\right)\frac{N-1}{N-p}$$

$$\boxed{\frac{1 - R^2_{adj}}{1 - R^2} = \frac{N-1}{N-p} \rightarrow \geq 1 \rightarrow R^2 \geq R^2_{adj}}$$

if $R^2 >> R^2_{adj}$ $\Longrightarrow$ Reduce the model

Can R^2_{adj} be negative?

When the model is **overfitted** and the R^2 is **low**, R^2_{adj} can be negative

$$R^2 = 1 - \frac{SSE}{SST} \geq 0 \quad \Rightarrow \quad R^2_{Adj} = 1 - \left(1 - R^2\right)\frac{N-1}{N-p}$$

$$R^2_{adj} < 0 \quad \Rightarrow \quad 1 - \left(1 - R^2\right)\frac{N-1}{N-p} < 0 \rightarrow \left(1 - R^2\right)\frac{N-1}{N-p} > 1 \rightarrow \left(1 - R^2\right) > \frac{N-p}{N-1}$$

$$1 - \frac{N-p}{N-1} > R^2 \quad \Rightarrow \quad R^2 < \frac{p-1}{N-1}$$

i.e. if p=6, N=11 if R^2<0.5 -> R^2_{adj} **is negative**

If the MODEL is Significant

MODEL IMPROVEMENT

– REDUCING (Less terms in the model)

– EXPANDING (More terms in the model)

Reducing: "Extra Sum of Squares" method

- Let's consider the **Full Model** $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

$$\boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \end{bmatrix} \quad \boldsymbol{\beta}_1 \rightarrow r*1 \quad \boldsymbol{\beta}_2 \rightarrow (p\text{-}r)*1 \qquad \mathbf{X} = [\mathbf{X}_1 \quad \mathbf{X}_2]$$

- **Full Model** $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$
- **Reduced Model** $\mathbf{y} = \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$
- **Extra Sum of Squares** method :

$$\boxed{H_0 : \boldsymbol{\beta}_1 = \mathbf{0} \quad \text{vs} \quad H_1 : \boldsymbol{\beta}_1 \neq \mathbf{0}}$$

$$\boxed{F_0 = \frac{\dfrac{SS^*_{REG}(F) - SS^*_{REG}(R)}{df^*_{REG}(F) - df^*_{REG}(R)}}{MS_E(F)} \sim F\left(df^*_{REG}(F) - df^*_{REG}(R), df_E(F)\right)}$$

$$df^*_{REG}(F) = p \quad df^*_{REG}(R) = (p - r) \qquad df_E(F) = N - p$$

Or, using a different expression

FULL MODEL

$$SS^*_{REG}\left(\hat{\boldsymbol{\beta}}\right) = \hat{\boldsymbol{\beta}}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{y} \qquad df^*_{REG}\left(\hat{\boldsymbol{\beta}}\right) = p$$

REDUCED MODEL

$$SS^*_{REG}\left(\hat{\boldsymbol{\beta}}_2\right) = \hat{\boldsymbol{\beta}}_2^T\mathbf{X}_2^T\mathbf{y} \qquad df^*_{REG}\left(\hat{\boldsymbol{\beta}}_2\right) = p - r$$

$$SS^*_{REG}(\hat{\boldsymbol{\beta}}) = SS^*_{REG}(\hat{\boldsymbol{\beta}}_2) + SS^*_{REG}(\hat{\boldsymbol{\beta}}_1 \mid \hat{\boldsymbol{\beta}}_2)$$

$$\boxed{\mathrm{H}_0: \boldsymbol{\beta}_1 = \mathbf{0} \quad \rightarrow \quad F_0 = \frac{SS^*_{REG}(\hat{\boldsymbol{\beta}}_1 \mid \hat{\boldsymbol{\beta}}_2)/r}{MS_E\left(F\right)} \sim F(r, N-p)}$$

PROOF
(simplified)

$$\mathbf{X} = \left[\mathbf{X}_1 \;\; \mathbf{X}_2 \right] \qquad \mathbf{X}^T = \begin{bmatrix} \mathbf{X}_1^T \\ \mathbf{X}_2^T \end{bmatrix} \rightarrow \mathbf{X}^T\mathbf{X} = \begin{bmatrix} \mathbf{X}_1^T\mathbf{X}_1 & \mathbf{X}_1^T\mathbf{X}_2 \\ \mathbf{X}_2^T\mathbf{X}_1 & \mathbf{X}_2^T\mathbf{X}_2 \end{bmatrix}$$

If the two parts of the model **<u>are</u>** **orthogonals**

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} \mathbf{X}_1^T\mathbf{X}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2^T\mathbf{X}_2 \end{bmatrix} \Rightarrow \left(\mathbf{X}^T\mathbf{X}\right)^{-1} = \begin{bmatrix} \left(\mathbf{X}_1^T\mathbf{X}_1\right)^{-1} & \mathbf{0} \\ \mathbf{0} & \left(\mathbf{X}_2^T\mathbf{X}_2\right)^{-1} \end{bmatrix}$$

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \left(\mathbf{X}_1^T\mathbf{X}_1\right)^{-1} & \mathbf{0} \\ \mathbf{0} & \left(\mathbf{X}_2^T\mathbf{X}_2\right)^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{X}_1^T\mathbf{y} \\ \mathbf{X}_2^T\mathbf{y} \end{bmatrix} = \begin{Bmatrix} \left(\mathbf{X}_1^T\mathbf{X}_1\right)^{-1} \mathbf{X}_1^T\mathbf{y} \\ \left(\mathbf{X}_2^T\mathbf{X}_2\right)^{-1} \mathbf{X}_2^T\mathbf{y} \end{Bmatrix} = \begin{Bmatrix} \hat{\boldsymbol{\beta}}_1 \\ \hat{\boldsymbol{\beta}}_2 \end{Bmatrix}$$

$$SS^*_{REG}\left(\hat{\boldsymbol{\beta}}\right) = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} = \left[\hat{\boldsymbol{\beta}}_1^T \;\; \hat{\boldsymbol{\beta}}_2^T\right] \begin{bmatrix} \mathbf{X}_1^T \\ \mathbf{X}_2^T \end{bmatrix} \mathbf{y} = \hat{\boldsymbol{\beta}}_1^T\mathbf{X}_1^T\mathbf{y} + \hat{\boldsymbol{\beta}}_2^T\mathbf{X}_2^T\mathbf{y} = SS^*_{REG}\left(\hat{\boldsymbol{\beta}}_1\right) + SS^*_{REG}\left(\hat{\boldsymbol{\beta}}_2\right)$$

$$\boxed{\begin{aligned} &SS^*_{REG}\left(\hat{\boldsymbol{\beta}}\right) = SS^*_{REG}\left(\hat{\boldsymbol{\beta}}_1\right) + SS^*_{REG}\left(\hat{\boldsymbol{\beta}}_2\right) \\ &p = r + (p - r) \end{aligned}}$$

$$H_0 : \boldsymbol{\beta}_1 = \mathbf{0} \rightarrow E\left(\hat{\boldsymbol{\beta}}_1\right) = \mathbf{0} \rightarrow \hat{\boldsymbol{\beta}}_1 \sim N\left(\mathbf{0}, \left(\mathbf{X}_1^T \mathbf{X}_1\right)^{-1} \sigma^2\right)$$

$$F_0 = \frac{SS^*_{REG}(\hat{\boldsymbol{\beta}}_1 \mid \hat{\boldsymbol{\beta}}_2)/r}{MS_E} = \frac{\left[SS^*_{REG}\left(\hat{\boldsymbol{\beta}}\right) - SS^*_{REG}(\hat{\boldsymbol{\beta}}_2)\right]/r}{MS_E} == \frac{SS^*_{REG}\left(\hat{\boldsymbol{\beta}}_1\right)/r}{SS_E/df_E}$$

$$= \frac{\hat{\boldsymbol{\beta}}_1^T \mathbf{X}_1^T \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 / r}{MS_E} = \frac{\hat{\boldsymbol{\beta}}_1^T \mathbf{X}_1^T \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 / r\sigma^2}{MS_E/\sigma^2} = \frac{\chi^2(r)/r}{\chi^2(df_E)/df_E} \sim F(r, df_E)$$

- In general the two parts are not **orthogonals** and we need a more complex proof

Model Coefficient Significance

- **Regression model** significance does not imply that every model parameter is significant
- We want to test each parameter
- We know that:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{y} \quad \begin{cases} \hat{\boldsymbol{\beta}} \text{ is Multivariate normal} \\ E\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta} \\ V\left(\hat{\boldsymbol{\beta}}\right) \equiv \boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}} = (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\sigma^2 \end{cases}$$

$\boxed{\text{We can build the } \textbf{Parameter Confidence Region}}$

- Multivariate normal probability density function

$$f\left(\hat{\boldsymbol{\beta}}\right)=\frac{1}{\left(2\pi\right)^{p/2}\left|\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}}\right|^{1/2}}\exp\left\{-\frac{1}{2}\left(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}\right)^{T}\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}}^{-1}\left(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}\right)\right\}$$

and $\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}}$ definite positive

- Confidence ellipsoid $\boxed{\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)^{T}\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}}^{-1}\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)\le\chi_{\alpha}^{2}(p)}$
- or $\boxed{\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)^{T}\left(\mathbf{X}^{T}\mathbf{X}\right)\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)\le\sigma^{2}\chi_{\alpha}^{2}(p)}$
- It is an ellipsoid centered in **beta_hat**
- The **semiaxis lenghts** are associated to the Eigenvalues of the matrix $\boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}}\rightarrow\sigma\sqrt{\lambda_{i}\chi_{\alpha}^{2}\left(p\right)}$
- The Axe Slopes are given by the Eigenvectors

- Usually we do not know σ^2 and we estimate it using MS_E with df_E degrees of freedom.
- The parameter confidence region becomes

$$\boxed{\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)^{\mathbf{T}}\left(\mathbf{X}^{\mathbf{T}}\mathbf{X}\right)\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)\leq pMS_E F_\alpha\left(p,df_E\right)}$$

where MS_E estimates σ^2 with df_E degrees of freedom

$$\text{Remark: } \hat{\boldsymbol{\Sigma}}_{\hat{\boldsymbol{\beta}}}=\left(\mathbf{X}^T\mathbf{X}\right)^{-1}MS_E$$

Proof

- We know that

$$\frac{\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)^T\left(\mathbf{X}^T\mathbf{X}\right)\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)}{\sigma^2}\sim\chi^2(p) \Longrightarrow \frac{\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)^T\left(\mathbf{X}^T\mathbf{X}\right)\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)}{p\sigma^2}\sim\frac{\chi^2(p)}{p}$$

- Moreover

$$\frac{SS_E}{\sigma^2}\sim\chi^2\left(df_E\right) \Longrightarrow \frac{MS_E}{\sigma^2}\sim\frac{\chi^2\left(df_E\right)}{df_E}$$

$$\frac{\dfrac{\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)^T\left(\mathbf{X}^T\mathbf{X}\right)\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)}{p\sigma^2}}{\dfrac{MS_E}{\sigma^2}}\sim\frac{\dfrac{\chi^2(p)}{p}}{\dfrac{\chi^2\left(df_E\right)}{df_E}}\sim F\left(p,df_E\right)$$

- Eventually

$$\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)^T\left(\mathbf{X}^T\mathbf{X}\right)\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)\sim pMS_EF\left(p,df_E\right)$$

Approximate method

Let's call c_{jj} the diagonal terms of $\left(\mathbf{X}^T\mathbf{X}\right)^{-1}$

We can state that

$$\hat{\beta}_j \sim N(\beta_j, c_{jj}\sigma^2) \rightarrow \frac{\hat{\beta}_j - \beta_j}{\sqrt{c_{jj}\sigma^2}} \sim \mathrm{N}(0,1) \qquad \frac{\hat{\beta}_j - \beta_j}{\sqrt{c_{jj}MS_E}} \sim t(N-p)$$

$$\text{where} \quad \sqrt{c_{jj}MS_E} \equiv se(\hat{\beta}_j) \rightarrow \text{Standard error of } \hat{\beta}_j$$

For a single model parameter

$$\boxed{\hat{\beta}_j - se\left(\hat{\beta}_j\right)t_{\alpha/2}(N-p) \le \beta_j \le \hat{\beta}_j + se\left(\hat{\beta}_j\right)t_{\alpha/2}(N-p)}$$

Or $\left|\hat{\beta}_j / se\left(\hat{\beta}_j\right)\right| > t_{\alpha/2}(N-p) \rightarrow \text{Reject } \beta_j = 0$

The approximate **Parameter Confidence Region** becomes a **Rectangular region** if we consider all model parameters

We can build ***p*** Confidence Intervals or make p test on the significance of model parameters using the Bonferroni's approach.

$$\left|\hat{\beta}_j / se\left(\hat{\beta}_j\right)\right| > t_{\alpha/2p}(N-p) \rightarrow \text{Reject } H_0 : \beta_j = 0$$

If you do not want to test the constant term β_0, substitute ***p*** with ***k=p-1***

- We could be interested to test

$$H_0 : \mathbf{a}^T \boldsymbol{\beta} = 0 \quad \text{vs} \quad H_1 : \mathbf{a}^T \boldsymbol{\beta} \neq 0$$

$$\mathbf{a}^T \hat{\boldsymbol{\beta}} \sim \text{Gaussian distribution}$$

$$E\left(\mathbf{a}^T \hat{\boldsymbol{\beta}}\right) = \mathbf{a}^T \boldsymbol{\beta} \text{ for unbiased estimates}$$

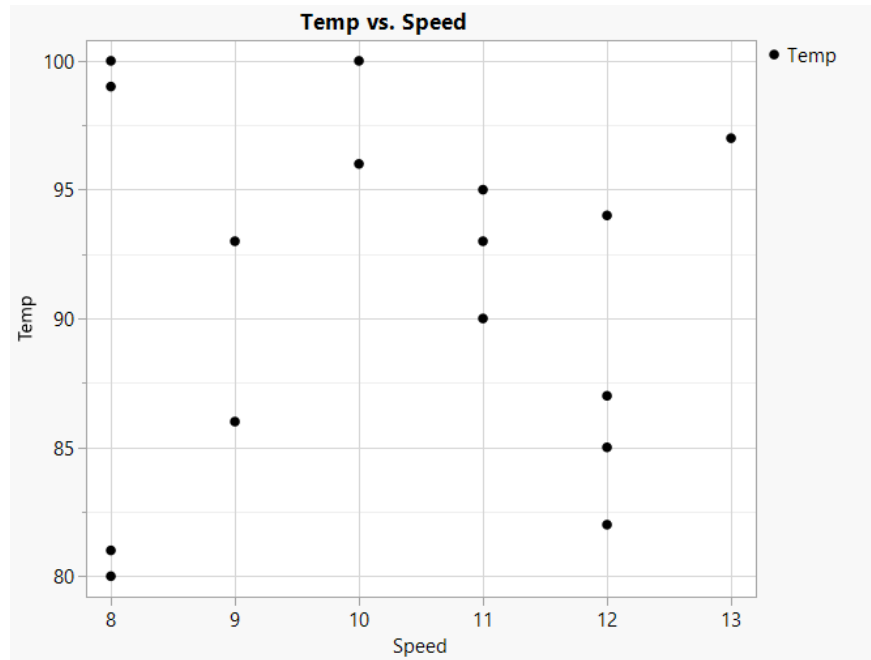
$$V\left(\mathbf{a}^T \hat{\boldsymbol{\beta}}\right) = \mathbf{a}^T V\left(\hat{\boldsymbol{\beta}}\right)\mathbf{a} = \mathbf{a}^T \left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{a}\sigma^2$$

$$\boxed{\left| \frac{\mathbf{a}^T \hat{\boldsymbol{\beta}}}{\sqrt{\mathbf{a}^T \left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{a} MS_E}} \right| > t_{\alpha/2}\left(df_E\right) \quad \Rightarrow \quad \text{Reject } H_0}$$

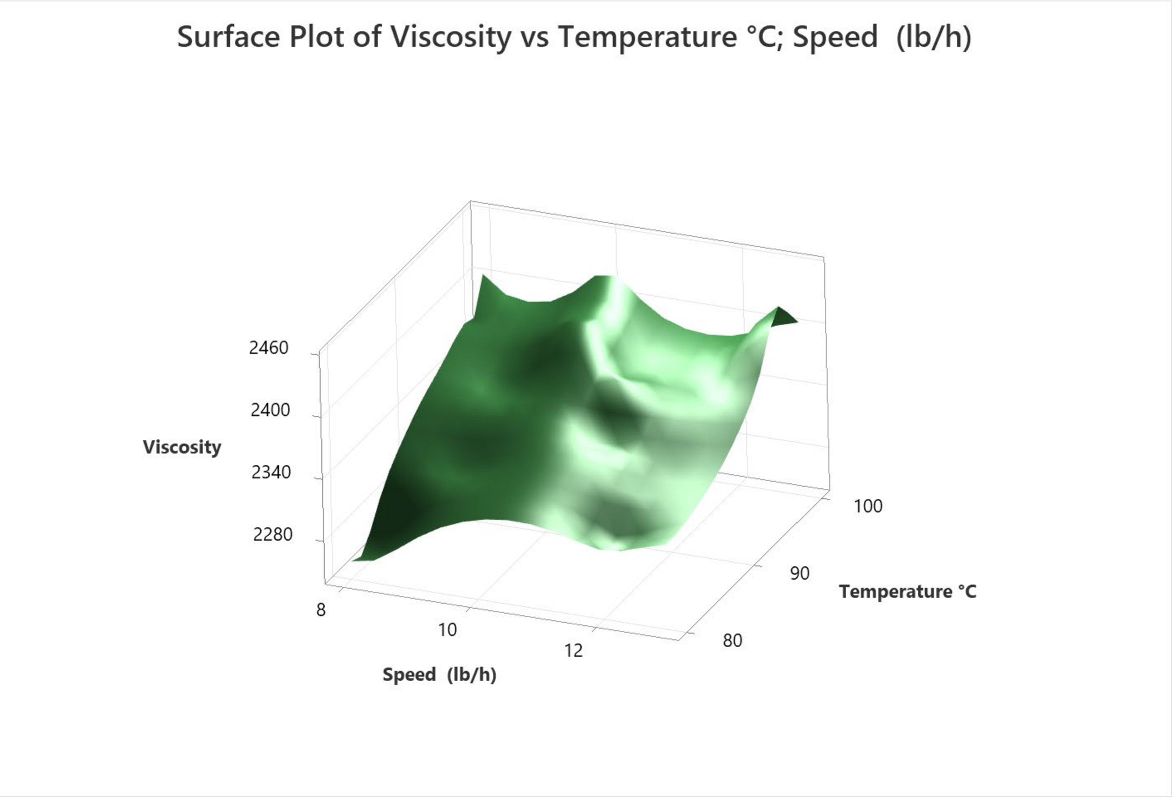
Example 10.1

We are interested in the effect of the **reaction temperature** and the **cathalyst feeding speed** on the **viscosity of a polymer**. We got 16 observations

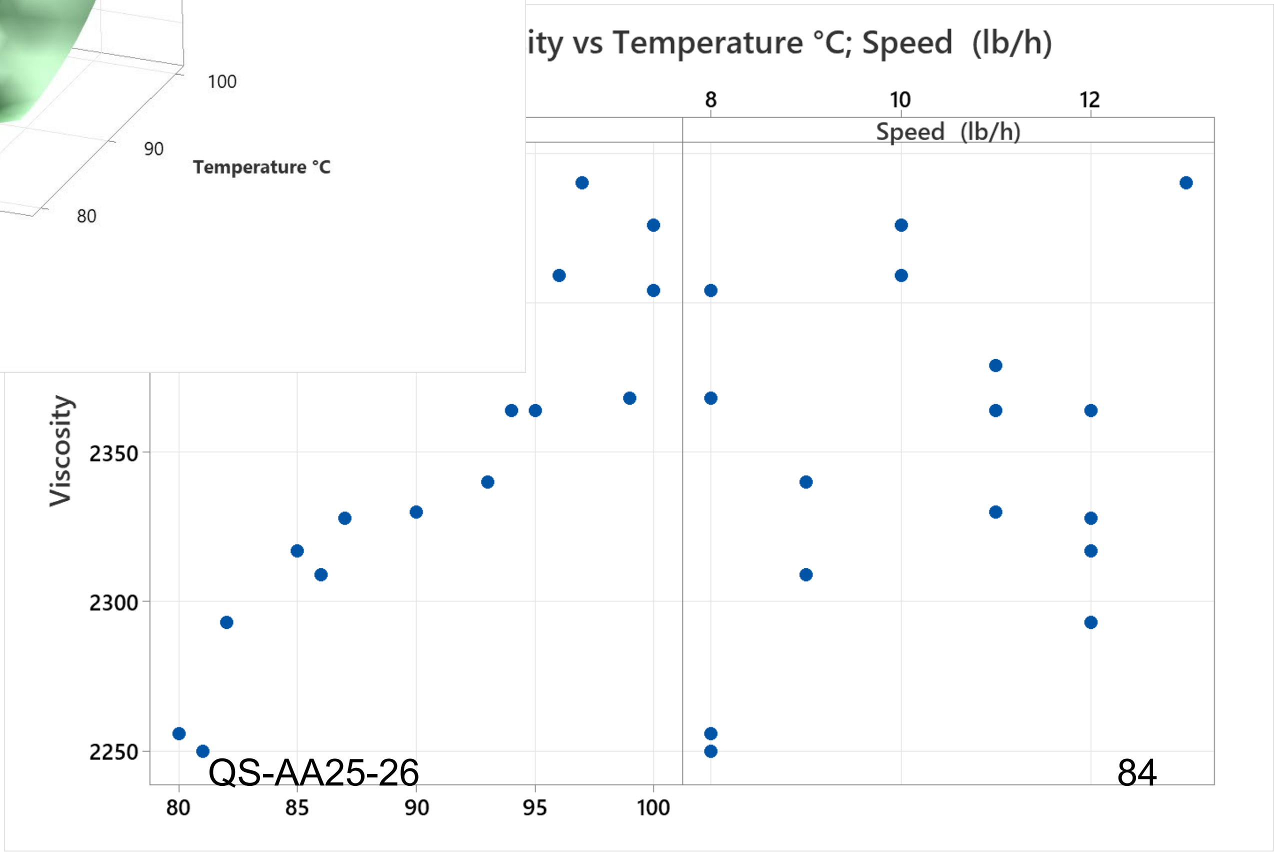
Observ.	Viscosity	Temp. (°C)	Speed (lb/h)
1	2256	80	8
2	2340	93	9
3	2426	100	10
4	2293	82	12
5	2330	90	11
6	2368	99	8
7	2250	81	8
8	2409	96	10
9	2364	94	12
10	2379	93	11
11	2440	97	13
12	2364	95	11
13	2404	100	8
14	2317	85	12
15	2309	86	9
16	2328	87	12



Data Graphs



Surface Plot of Viscosity vs Temperature °C; Speed (lb/h)
2460
2400
2340
2280
Viscosity
8
10
12
Speed (lb/h)
80
90
100
Temperature °C



ity vs Temperature °C; Speed (lb/h)
8
10
12
Speed (lb/h)
Viscosity
2350
2300
2250
80
85
90
95
100

Model guess

Let us assume an additive model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \qquad \text{with } \varepsilon \sim NID(0,\sigma^2)$$

Write the $\mathbf{y}$ (16x1) vector

Write the $\mathbf{X}$ (16x3) Matrix

X^TX (3x3) Matrix

16.0	1458.0	164.0
1458.0	133560.0	14946.0
164.0	14946.0	1726.0

$(X^TX)^{-1}$ Matrix

14.176004	-0.129746	-0.22345
-0.129746	$1.429184*10^{-3}$	$-4.763947*10^{-5}$
-0.223453	$-4.763947*10^{-5}$	$2.222381*10^{-2}$

$\mathbf{X}^T\mathbf{y}$ (3x16)(16x1)=(3x1)

37577.0

3429550.0

38556.0

Now we can compute the estimates of the model parameters

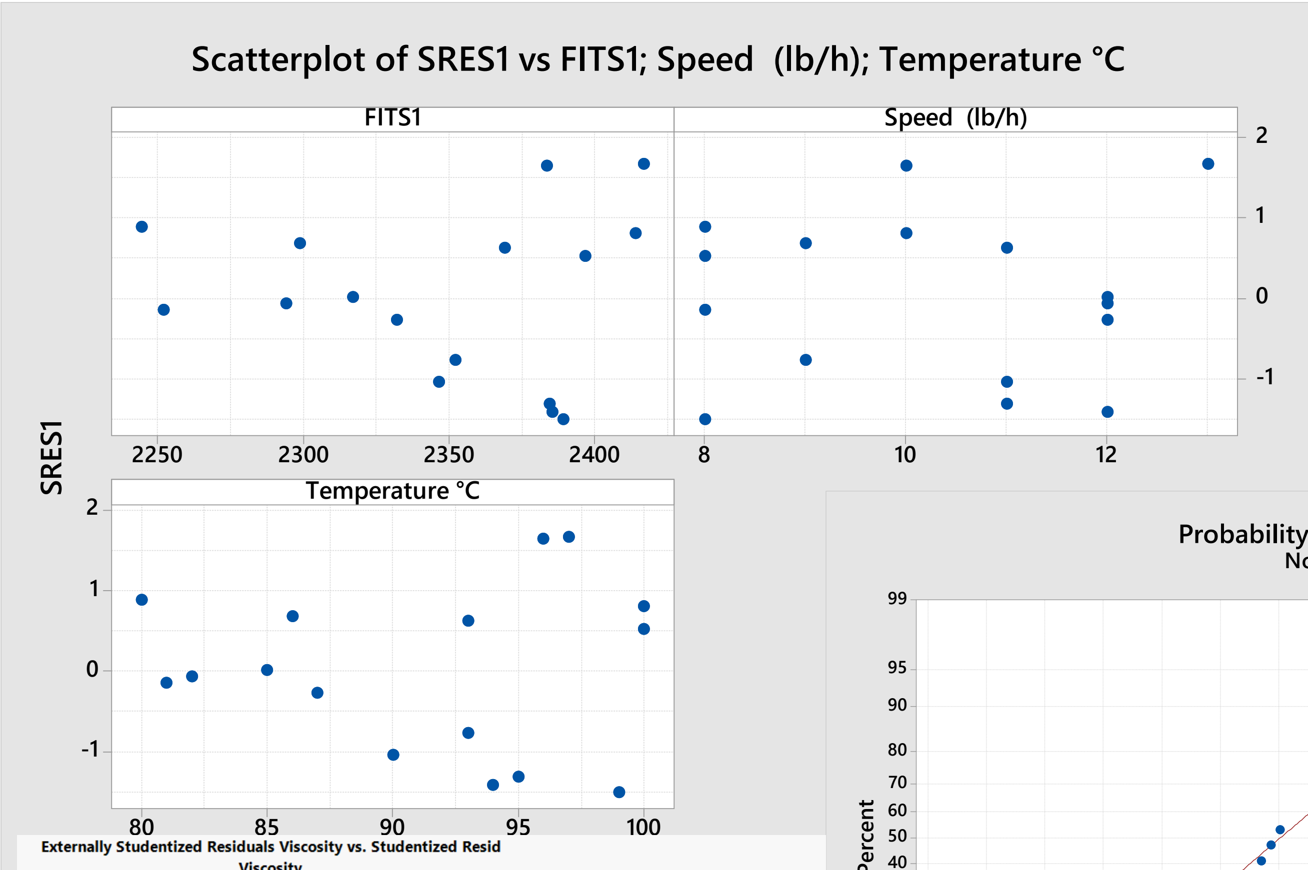
$$\hat{\beta}_0 = b_0 = 1566.078$$

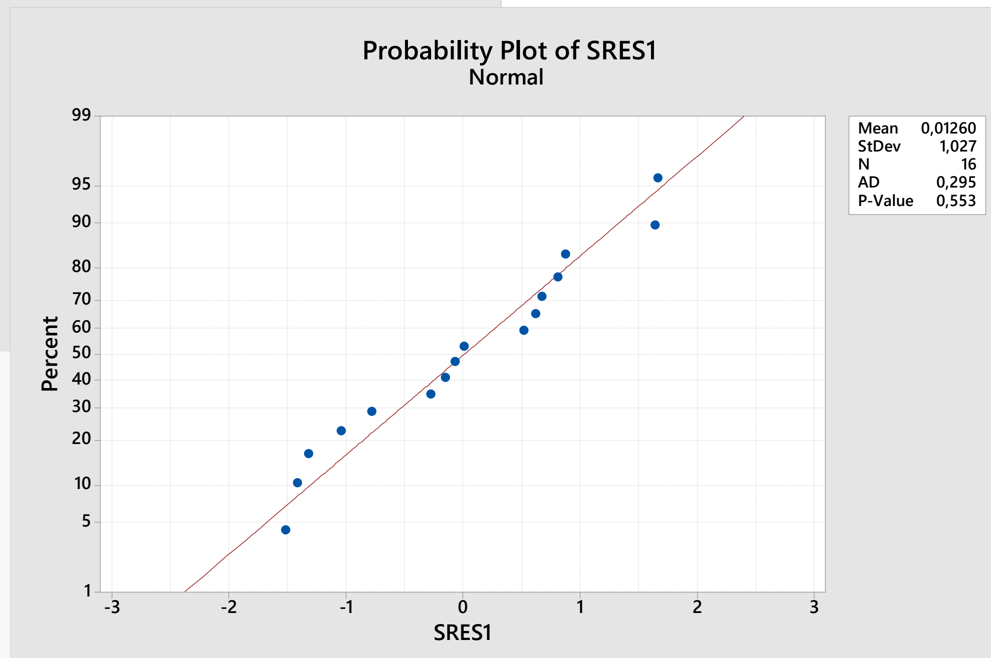
$$\hat{\beta}_1 = b_1 = 7.62129$$

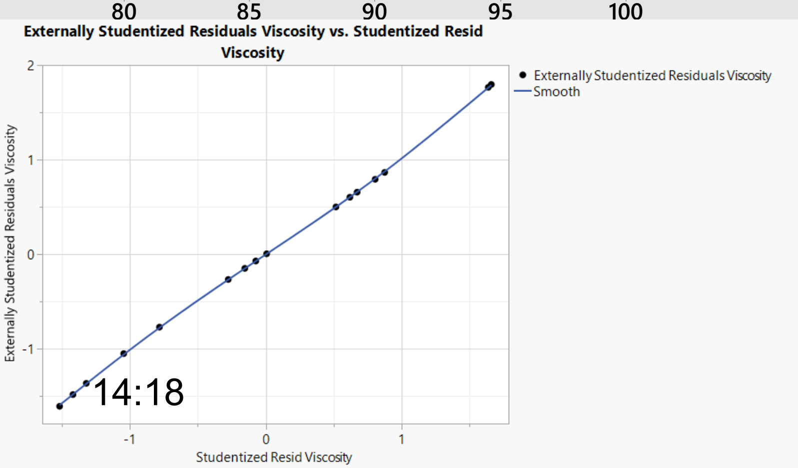
$$\hat{\beta}_2 = b_2 = 8.58485$$

And compute Fits and StdResiduals

Residual checking

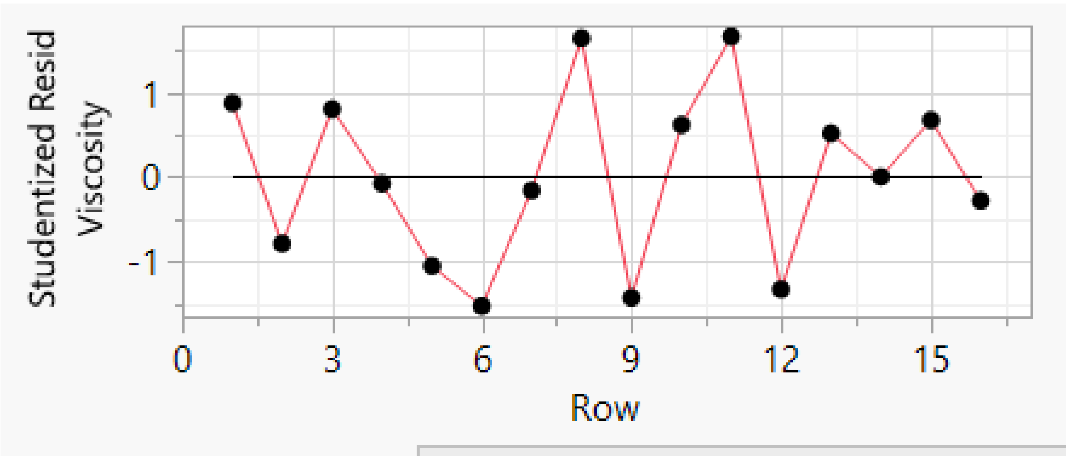






14:18

- Let us assume that the observations are in a temporal sequence
- Plot the data according to their order



Autocorrelation Test

Time Series Basic Diagnostics

Lag	AutoCorr	-.8-.6-.4-.2 0 .2 .4 .6 .8	Ljung-Box Q	p-Value
0	1.0000		.	.
1	-0.3130		1.8806	0.1703
2	-0.1918		2.6377	0.2674
3	0.2399		3.9122	0.2711
4	-0.0680		4.0233	0.4029
5	-0.1241		4.4263	0.4898

ANOVA table and σ^2 estimate

Source	SS Formula	SS Value	*df*	MS	F_0
Regression (corrected)	$\hat{\boldsymbol{\beta}}^T \mathbf{X}^T \mathbf{y} - \frac{y_{\bullet}^2}{N}$	44157	2	22079	**82.5**
Residual Error	$\mathbf{y}^T \mathbf{y} - \hat{\boldsymbol{\beta}}^T \mathbf{X}^T \mathbf{y}$	3479	13	267,6	
Total (corrected)	$\mathbf{y}^T \mathbf{y} - \frac{y_{\bullet}^2}{N}$	47636	15		

$F_{0.05}(2,13) = 3.81$

$p = 0.00$

$$\frac{y_{\bullet}^2}{N} = \frac{(37577)^2}{16} = 88251933$$

$$\mathbf{y}^T \mathbf{y} = 88299569$$

$$\boxed{\begin{aligned} \hat{\sigma}^2 &= MS_E = 267.6 \\ \hat{\sigma} &= 16.3 \end{aligned}}$$

$$\boldsymbol{\beta}^T \mathbf{X}^T \mathbf{y} = \begin{bmatrix} 1566.078 & 7.62129 & 8.58485 \end{bmatrix} \begin{bmatrix} 37577.0 \\ 3429550.0 \\ 385562.0 \end{bmatrix} = 88296090$$

- Compute **R²** $$R^2 = \frac{SS_{REG}}{SS_{TOT}} = 1 - \frac{SS_E}{SS_{TOT}} = 1 - \frac{3479}{47636} \rightarrow 92.7\%$$

- Compute **R²adj** $$R^2_{adj} = 1 - \frac{SS_E / (N-p)}{SS_T / (N-1)} = 1 - \frac{3479/(16-3)}{47636/(16-1)} \rightarrow 91.6\%$$

Compute the parameter significance t-value

$$se(\hat{\beta}_0) = \sqrt{c_{00}\hat{\sigma}^2} = \sqrt{14.176 * 267.6} = 61.59 \qquad \hat{\beta}_0 / se(\hat{\beta}_0) = 25.43$$

$$se(\hat{\beta}_1) = \sqrt{c_{11}\hat{\sigma}^2} = \sqrt{1.429184 * 10^{-3} * 267.6} = 0.618 \qquad \hat{\beta}_1 / se(\hat{\beta}_1) = 12.32$$

$$se(\hat{\beta}_2) = \sqrt{c_{22}\hat{\sigma}^2} = \sqrt{2.222381 * 10^{-2} * 267.6} = 2.439 \qquad \hat{\beta}_2 / se(\hat{\beta}_2) = 3.52$$

these values must be compared to $t_{\alpha/2}(13)$ o $t_{\alpha/2p}(13)$

$$t_{2.5\%}(13) = 2.1604 \qquad t_{0.\overline{3}\%}(13) = 2.7459$$

- Build the Confidence Interval (CI) for $\boldsymbol{\beta_2}$ with Confidence coefficient 95%

$$df_E = N - p = 16 - 3 = 13 \quad \hat{\sigma}^2 = 267.6 \quad c_{22} = 0.0222$$

$$se(\hat{\beta}_2) = 2.439 \quad t_{0.025}(13) = 2.16 \quad \hat{\beta}_2 = 8.585$$

$$\hat{\beta}_2 - t_{0.025}(13)se(\hat{\beta}_2) \le \beta_2 \le \hat{\beta}_2 + t_{0.025}(13)se(\hat{\beta}_2)$$

$$8.585 - 2.16 \cdot 2.439 \le \beta_2 \le 8.585 + 2.16 \cdot 2.439$$

$$\boxed{3.32 \le \beta_2 \le 13.85}$$

JMP v17 output

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	2	44157.087	22078.5	82.5046
Error	13	3478.851	267.6	Prob > F
C. Total	15	47635.938		<.0001*

Summary of Fit

RSquare	0.92697
RSquare Adj	0.915735
Root Mean Square Error	16.3586
Mean of Response	2348.563
Observations (or Sum Wgts)	16

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Temp	1	1	40641.418	151.8715	<.0001*
Speed	1	1	3316.244	12.3924	0.0038*

Note:
SS(Temp|Cost)+SS(Speed|Cost)=
40641.418+3316.244=43957.662
Which is different from 44157.087

Why??

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|	VIF
Intercept	1566.0778	61.59184	25.43	<.0001*	.
Temp	7.6212901	0.61843	12.32	<.0001*	1.0000715
Speed	8.5848459	2.438684	3.52	0.0038*	1.0000715

Non orthogonal plan

- JMP computes the marginal SS associated with the regression term of interest
- Let us suppose a model with p parameters

$$SS_{REG}\left(F\right)=SS\left(\hat{\boldsymbol{\beta}}\mid\hat{\beta}_0\right)\quad \text{where } \hat{\boldsymbol{\beta}} \text{ is a vector with dimension } \left(p-1\right)*1$$

$$\text{Marginal } SS\left(\hat{\beta}_1\right)\rightarrow SS_{REG}\left(\hat{\beta}_1\mid\hat{\boldsymbol{\beta}}_{(1)},\hat{\beta}_0\right)=SS_{REG}\left(F\right)-SS_{REG}\left(\hat{\boldsymbol{\beta}}_{(1)}\mid\hat{\beta}_0\right)$$

$$SS_{REG}\left(\beta_i\mid\hat{\boldsymbol{\beta}}_{(i)},\hat{\beta}_0\right)=SS_{REG}\left(F\right)-SS_{REG}\left(\hat{\boldsymbol{\beta}}_{(i)}\mid\hat{\beta}_0\right)$$

$$SS_{REG}\left(F\mid\hat{\beta}_0\right)\neq\sum_{i=1,k}SS_{REG}\left(\beta_i\mid\hat{\boldsymbol{\beta}}_{(i)},\hat{\beta}_0\right)$$

- Let us compute the **studentized residual** for the 8th observation

$$e_8 = 25.43 \quad \hat{\sigma}^2 \equiv MS_E = 268 \quad h_{88} = 0.09797$$

$$SRES_8 = \frac{25.43}{\sqrt{268\left(1-0.09797\right)}} = 1.6356 \rightarrow 1.6368 \text{ (by JMP)}$$

Example 10.6 (continuation of example 10.1)

Let us consider the example 10.1 data and test the significance of the **speed** factor

$$H_0 : \beta_2 = 0$$

$$H_1 : \beta_2 \neq 0$$

$$SS^*_{REG}(\hat{\beta}_2 / \hat{\beta}_1, \hat{\beta}_0) = SS^*_{REG}(\hat{\beta}_2, \hat{\beta}_1, \hat{\beta}_0) - SS^*_{REG}(\hat{\beta}_1, \hat{\beta}_0)$$

$$= SS_{REG}(\hat{\beta}_2, \hat{\beta}_1 / \hat{\beta}_0) - SS_{REG}(\hat{\beta}_1 / \hat{\beta}_0)$$

$$SS_{REG}(\hat{\beta}_2, \hat{\beta}_1 / \hat{\beta}_0) = 44157.1 \text{ with } 2\ df$$

$$SS_{REG}(\hat{\beta}_1 / \hat{\beta}_0) = 40840.8 \qquad \text{with } 1\ df$$

$$SS_{REG}(\hat{\beta}_2 / \hat{\beta}_1, \hat{\beta}_0) = 44157.1 - 40840.8 = 3316.3 \text{ with } 1\ df$$

$$MS_E = 267.604 \text{ with } 13 \; df$$

$$F_0 = \frac{3316.3}{267.604} = 12.392 \quad \text{to compare with } F_{0.05}(1,13) = 4.67$$

$\boxed{p\text{-}value = 0.003765}$

in this particular case (the first parameter of F is 1)

you may compute

$$se(\hat{\beta}_2) = \sqrt{0.0222 * 266.9} = 2.434$$

$$\hat{\beta}_2 = 8.58485 \qquad F_{0.05}(1,13) = t^2_{0.025}(13)$$

$$\left| \frac{\hat{\beta}_2}{se(\hat{\beta}_2)} \right| = 3.53 \text{ to compare with } t_{0.025}(13) = 2.16$$